Behavioral Inhibition as a Probabilistic Anxiety Marker

Early behavioral inhibition is associated with later social-anxiety risk at the group level; it is neither necessary nor sufficient, and stranger wariness is not a diagnosis.

MODERATE-GROUP-RISK; LOW-INDIVIDUAL-FORECAST | STATUS: DRAFT | DOMAIN: EMOTION, TEMPERAMENT | REPLICATION: VE

WHAT WE KNOW (Narrowest Supported Conclusion)

Some toddlers consistently withdraw from unfamiliar people, objects, or settings. That pattern—behavioral inhibition—is a temperament risk marker for later anxiety, especially social anxiety. Clauss & Blackford (2012) meta-analyzed longitudinal studies and found elevated odds of later social-anxiety disorder among inhibited children. The association is probabilistic: most inhibited toddlers do not become clinically anxious, and anxiety occurs without an inhibited history.

EVIDENCE AND CAUSAL ASSESSMENT

The identification is longitudinal prediction, not an RCT of temperament. Selective samples, shared informants, and reduced social opportunity can inflate estimates. Intervention trials in inhibited preschoolers (outside this card's primary ID) suggest anxiety risk can move more than observer-rated inhibition itself. Adjacent Chapter 8: stranger wariness ≠ insecure attachment. Adjacent: temperament-reactivity-not-destiny.md is the broader dimensional claim.

EFFECT SIZE AND PERSISTENCE

Odds ratios in the meta-analysis are meaningful at the group level and heterogeneous across anxiety disorders. Persistence is not identity: inhibition can recede; later anxiety can emerge without it.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: untested-in-GI (behavioral inhibition is a moderately heritable phenotype with shared genetic liability for social anxiety; individual risk forecasts are not deterministic; quantitative attenuation in twin/adoption designs not formally parameterized for this specific longitudinal link).

Shared method variance (parent report of both inhibition and later anxiety) is a live threat. Reduced exposure to novelty can maintain both the marker and the outcome.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Normative stranger anxiety, separation distress, or caution is not clinical anxiety. Early inhibition does not fix a child's future. Forcing exposure as character training, or treating a wary toddler as an attachment failure, does not follow.

WHAT ACTION IS REASONABLY JUSTIFIED

Distinguish expected fear from persistent impairment across settings. Proportionate assessment when functioning is affected is consistent with the evidence. Destiny ranking is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Clauss & Blackford (2012), Journal of the American Academy of Child and Adolescent Psychiatry

Breastfeeding and Cognitive Development

PROBIT RCT and sibling-comparison evidence indicate a small, verbal-concentrated, fading causal effect of breastfeeding on IQ.

GRADE: MODERATE | STATUS: DRAFT | DOMAIN: FEEDING, COGNITION | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Observational studies consistently show a 3-4 IQ-point advantage for breastfed children, but the strongest causal designs (PROBIT cluster-randomized trial follow-up, sibling comparisons) indicate the long-term causal effect for full-term infants is small, concentrated in verbal ability, and may approach zero by adolescence.

EVIDENCE AND CAUSAL ASSESSMENT

- PROBIT (Belarus, N=17,046 randomized at the hospital/cluster level) found infants in the breastfeeding-promotion arm scored 5.9 points higher full-scale IQ (not significant) and 7.5 points higher verbal IQ (significant) at age 6.5.

- 16-year PROBIT follow-up (79.5% retention, N=13,557) found no significant difference in overall neurocognitive function; only verbal function remained significantly higher (+1.4 points).

- Sibling-comparison studies (Der 2006; Colen & Ramey 2014) using within-family designs to remove shared confounders found no significant cognitive differences between breastfed and non-breastfed siblings.

EFFECT SIZE AND PERSISTENCE

Randomized estimate at age 6.5: +5.9 full-scale IQ (ns), +7.5 verbal IQ (significant). At age 16: overall effect non-significant; verbal effect fades to +1.4 points. Sibling designs: null. Best summary estimate for full-term, adequately nourished infants: 0-2 IQ points, likely concentrated in verbal domain, with fade-out over adolescence.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: largely confounded (>50%) (observational IQ gains largely attenuate in sibling-comparison designs and PROBIT trials).

Partially-to-largely confounded relative to observational estimates: PROBIT's cluster-randomized design removes maternal-choice confounding, yet the effect that remains after 16 years is far smaller than initial observational estimates (~3-4 points) and disappears in sibling comparisons. Findings may not generalize to preterm or malnourished populations.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not support claims that breastfeeding produces large or lifelong IQ gains, nor that formula-feeding meaningfully harms cognitive development in full-term, well-nourished populations.

WHAT ACTION IS REASONABLY JUSTIFIED

Breastfeeding promotion is justifiable on other well-established grounds (infection reduction, maternal health, cost), not primarily on cognitive-enhancement grounds for full-term infants in high-resource settings.

VERIFIED SOURCES & DOI IDENTIFIERS

- Kramer et al. (2008), Arch Gen Psychiatry, PROBIT IQ at 6.5 years
- Yang et al. (2018), PLOS Medicine, 16-year PROBIT follow-up
- Der, Batty & Deary (2006), BMJ, sibling pairs and meta-analysis
- Colen & Ramey (2014), Soc Sci Med, sibling comparisons
- Evenhouse & Reilly (2005), sibling comparisons (catalog ID sibling-comparison-2006)

Childhood Self-Control Is a Population Gradient, Not a Sentence

In the Dunedin cohort, childhood self-control predicts later health, wealth, and public-safety outcomes at the group level; improvement changes standing, and this is not a marshmallow-test replication.

: MODERATE-GRADIENT; LOW-INDIVIDUAL-FORECAST | STATUS: DRAFT | DOMAIN: REGULATION, LIFE-COURSE | REPLICATION: VE

WHAT WE KNOW (Narrowest Supported Conclusion)

Moffitt et al. (2011) followed the Dunedin birth cohort: a composite of childhood self-control (ages 3–11) predicted adult health, wealth, and public-safety outcomes, including in sibling comparisons. Children whose self-control improved across childhood had better later outcomes than their early standing implied. The object is a population gradient, not a single laboratory delay trial at age four.

EVIDENCE AND CAUSAL ASSESSMENT

Single-cohort longitudinal design with sibling checks. Not genetically informed attenuation in the CONTRIBUTING sense on this card. The Watts 2018 NICHD replication (self-regulation-delay-of-gratification.md) already showed that the classic marshmallow school-success story was mostly confounded; this card does not reopen that identification. Adjacent Chapter 11 cards describe toddler inhibition as a task family, not a life-course law.

EFFECT SIZE AND PERSISTENCE

Effects are described as a gradient across the distribution, not only the clinical tail. They are modest relative to a life, and they can move when self-control changes.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: untested-in-GI (childhood self-control is moderately heritable and shares genetic variance with socioeconomic gradients; specific genetically informed attenuation studies on Dunedin gradient estimates remain unparameterized on this card).

Family resources, IQ, and later environments remain entangled. Childhood self-control here is a multi-age composite, not a 36-month snapshot.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Low self-control at three does not doom a child. This is not a marshmallow-test replication. It does not justify ranking toddlers or converting ordinary life into a self-control curriculum.

WHAT ACTION IS REASONABLY JUSTIFIED

Scaffold regulation and reduce cumulative risk at the population level. Individual destiny language does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Moffitt et al. (2011), PNAS

Committed Compliance Is Not the Same as Obedience

Kochanska's observational program distinguishes committed (willing) compliance from situational compliance; the former is more closely linked to later internalization, and neither is a moral verdict or a parenting score.

RADE: MODERATE-DISTINCTION; LOW-TECHNIQUE | STATUS: DRAFT | DOMAIN: BEHAVIOR, REGULATION | REPLICATION: VERIFIE

WHAT WE KNOW (Narrowest Supported Conclusion)

A toddler can follow a request because an adult is standing there, or because the child has taken the request as their own. Kochanska (1995) traced committed compliance—willing, self-regulated cooperation with a caregiver's agenda—as more closely related to later internalization of rules than situational, pressure-dependent compliance. Understanding a "no" and being able to inhibit an act are not the same achievement.

EVIDENCE AND CAUSAL ASSESSMENT

Home and laboratory observations in modest, culturally bounded samples. Longitudinal associations are compatible with child characteristics, relationship quality, and bidirectional effects. This is not an RCT of a discipline package. Adjacent Chapter 11: inhibitory-control-toddler.md is task suppression, not this relational distinction. Sleep and attachment-intervention cards are not restated.

EFFECT SIZE AND PERSISTENCE

The relevant object is not "obedience" as a single score. Context-specific stability is more defensible than a durable compliance trait. "Do" and "don't" contexts can diverge.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: untested-in-GI (mutually responsive orientation and compliance reflect both evocative child temperament and parental scaffolding; observational laboratory dyads lack genetically informed twin/adoption controls).

Language comprehension, fatigue, hearing, and cultural expectations about deference can be mistaken for defiance. Samples are often maternal and Western.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Immediate compliance is not understanding, cooperation, conscience, or moral development. Noncompliance is not automatically a behavior problem. A failed "no" is not deficient parenting. Opposition can express agency, limited regulation, or communication difficulty.

WHAT ACTION IS REASONABLY JUSTIFIED

Treat "no" as communication that has to be cognitively and emotionally usable. A universal compliance script does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Kochanska (1995), Child Development

Conversational Turns Predict Later Language Better Than Raw Word Counts

In LENA recordings, adult–child conversational turns at 18–24 months predict school-age language and IQ after SES control more robustly than adult word counts; 18-month expressive vocabulary norms (CDI median ~50 words, range 10-170+) show vast typical variance.

GRADE: MODERATE-PREDICTION; LOW-CAUSATION | STATUS: DRAFT | DOMAIN: LANGUAGE | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Language quantity, normative milestone timing, and interactional quality are distinct developmental dimensions. Parents often ask how many words should an 18-month-old say: normative standardization on the MacArthur-Bates Communicative Development Inventories (CDI; Fenson et al. 2007) establishes that the 50th percentile for an 18-month-old child is approximately 50 spoken words, but typical variability spans from 10 words to over 170 words. Automated daylong LENA recordings demonstrate that adult–child conversational turns at 18–24 months account for 14–27% of variance in IQ, verbal comprehension, and vocabulary a decade later after socioeconomic status is controlled, whereas raw adult word counts in the same window are weakened by covariate controls.

EVIDENCE AND CAUSAL ASSESSMENT

- Fenson et al. (2007), MacArthur-Bates Communicative Development Inventories (CDI): Normative tracking across thousands of toddlers shows that expressive language accelerates non-linearly around 18 months. An 18-month-old child saying 15–20 words is well within the typical distribution if receptive language, gesture pointing, and communicative reciprocity are intact. Asking how many words an 18-month-old child should say requires looking at the wide distribution (10th percentile ~ 10 words, median ~ 50 words, 90th percentile ~ 170+ words) rather than a single cutoff.

- Gilkerson et al. (2018), Pediatrics: 146 children with monthly daylong recordings; follow-up at 9–14 years (WISC-V, PPVT, EVT). Observational, automated measurement, SES covariate. Turn-taking was substantially more predictive than raw adult word counts. Child-to-adult reciprocal exchange and shared communicative engagement drive the statistical link.

EFFECT SIZE AND PERSISTENCE

Reported 14–27% variance at a decade is a prediction interval in one cohort, not an artificial intervention effect. Normative word counts at 18 months show modest rank-order stability that strengthens as grammar emerges after 24 months.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: partially confounded (20–50%) (conversational turn-taking and early vocabulary share unmeasured genetic variance with parental verbal ability and child communicative drive; partially attenuated when family background and parental education are controlled).

LENA classification error; who talks more may already be the more verbal child; SES residual confounding; clinical late bloomers who catch up by age 3 without intervention.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean an 18-month-old child saying fewer than 50 words has a cognitive deficit or that parents should drill flashcards; the typical range at 18 months is wide (10 to 170+ words). Does not mean families with fewer recorded words care less, or that a missed month of conversation writes a child's future.

WHAT ACTION IS REASONABLY JUSTIFIED

Support time and natural conditions for contingent reciprocal conversation and shared reading. If an 18-month-old toddler says fewer than 10 words, does not respond to their name, or does not use pointing/gestures, a standard pediatric evaluation and hearing assessment is reasonably justified.

VERIFIED SOURCES & DOI IDENTIFIERS

- Gilkerson et al. (2018), Pediatrics (PMC6192025)

- Fenson et al. (2007), MacArthur-Bates Communicative Development Inventories: User's Guide and Technical Manual (2nd ed.), Brookes Publishing

Deferred Imitation After a Delay

Fourteen-month-olds can reproduce novel multi-step actions on objects after delays of up to one week without prior motor practice; this establishes preverbal recall memory, not neonatal imitation or an accelerated copying curriculum.

H-FOURTEEN-MONTH-ONE-WEEK; LOW-NEONATE-AND-TRANSFER | STATUS: STABLE | DOMAIN: LEARNING, MEMORY | REPLICATIO

WHAT WE KNOW (Narrowest Supported Conclusion)

Infants possess the capacity to encode an observed action sequence and retrieve that representation to guide motor action after a substantial temporal delay without intermediate motor practice. In controlled laboratory experiments, 14-month-old infants who observe an adult perform novel actions on unfamiliar objects (such as touching a forehead to a light panel or manipulating multi-part toys) reproduce those specific target actions after a 24-hour to one-week delay significantly more often than baseline controls who were exposed to the objects without the target demonstration.

EVIDENCE AND CAUSAL ASSESSMENT

Meltzoff (1988), Developmental Psychology, conducted a randomized controlled experiment with 14-month-old infants tested across a one-week delay. Infants observed an adult demonstrate six novel object actions (e.g., illuminating a light box with a forehead touch, assembling a dumbbell toy). After one week without physical contact with the objects, experimental infants produced significantly more target behaviors than baseline controls who experienced identical exposure time with the objects but without demonstrations, proving long-term preverbal declarative memory.

Barr, Dowden & Hayne (1996), Infant Behavior and Development, independently replicated and extended the developmental timeline of deferred imitation across 6-, 12-, 18-, and 24-month-olds across three experiments. At a 24-hour delay, infants aged 12 to 24 months robustly imitated multi-step puppet actions without prior practice; 6-month-olds demonstrated deferred imitation when given doubled demonstration exposure, establishing that deferred imitation is an indexing methodology for declarative memory maturation.

The causal identification is experimental: demonstration versus control presentation isolates memory for observed action from spontaneous exploratory manipulation.

EFFECT SIZE AND PERSISTENCE

In controlled trials, the effect size for target action reproduction following demonstration versus control exposure is large (Cohen's d > 0.80). However, the representation is context-bound: changing the visual features of the testing environment or altered toy cues significantly attenuates retrieval in younger infants. Performance indexes specific episodic recall, not an enduring general "imitation IQ."

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (laboratory operant modeling and recall delay, not an observational population phenotype association).

Motor competence, infant attention during demonstration, and stranger anxiety in the laboratory introduce within-subject measurement variance. Failure to imitate in the test session can reflect transient performance deficits (reluctance to touch unfamiliar items) rather than a failure of underlying memory.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean newborns possess innate mirror-like imitation of adult facial gestures as a settled fact. Does not mean toddlers copy every action they see or lack individual agency. Does not license commercial programs encouraging parents to drill infants on structured imitation tasks.

WHAT ACTION IS REASONABLY JUSTIFIED

Caregivers can expect that toddlers naturally watch, remember, and attempt to re-enact observed adult behaviors across everyday routines. Providing safe opportunities to explore household objects and tools is valuable, while formal imitation drills and worries about uncopied demonstrations are unnecessary.

VERIFIED SOURCES & DOI IDENTIFIERS

- Meltzoff (1988), Developmental Psychology (DOI: 10.1037/0012-1649.24.4.470 (https://doi.org/10.1037/0012-1649.24.4.470))

- Barr, Dowden & Hayne (1996), Infant Behavior and Development (DOI: 10.1016/s0163-6383(96)90015-6 (https://doi.org/10.1016/s0163-6383(96)90015-6))

Mode of Delivery and Neonatal Physiological Transition

Cesarean delivery alters immediate neonatal lung fluid clearance and neuroendocrine stress surges; observational associations with long-term neurodevelopment, autism, and cognitive traits attenuate to null under sibling-comparison designs.

HIGH-IMMEDIATE-PHYSIOLOGY; LOW-LONG-TERM-CAUSATION | STATUS: DRAFT | DOMAIN: BIRTH, NEONATAL | REPLICATION: VE

WHAT WE KNOW (Narrowest Supported Conclusion)

Mode of delivery affects the immediate physiological events of birth. Labor contractions and the accompanying surges of fetal catecholamines and glucocorticoids stimulate epithelial sodium channels (ENaC) in alveolar cells, shifting the fetal lung from fluid secretion to active fluid reabsorption. Cesarean delivery without labor bypasses this endocrine surge, resulting in higher retained lung liquid and elevated rates of transient tachypnea of the newborn (TTN). Similarly, neonatal physiological transition encompasses hepatic bilirubin clearance: transient neonatal jaundice affects over 60 percent of term newborns. Hospital blue-light phototherapy safely photoisomerizes unconjugated bilirubin into water-soluble lumirubin without long-term neurological sequelae. Furthermore, speculative procedures such as vaginal seeding following cesarean delivery (c-section)—swabbing operative newborns with maternal vaginal microbes to simulate birth canal colonization—currently lack randomized clinical trial evidence of lasting immunological or metabolic benefit. In contrast, observational associations between cesarean delivery and long-term child cognitive, behavioral, and neurodevelopmental outcomes (such as autism or ADHD) reflect indication confounding and family-level background rather than a causal effect of delivery mode itself.

EVIDENCE AND CAUSAL ASSESSMENT

Jain & Dudell (2006), Seminars in Perinatology, synthesizes the respiratory transition across delivery modes, detailing how the absence of labor-induced mechanical compression and catecholamine release delays alveolar fluid clearance, resulting in a higher clinical incidence of transient respiratory distress in scheduled cesareans.

Curran et al. (2015), JAMA Psychiatry, analyzed a Swedish population-based cohort of 2,697,315 singleton births, including 26,822 sibling pairs discordant for delivery mode. While conventional population-level models showed a modest observational increase in autism spectrum disorder following elective cesarean delivery (hazard ratio ~1.2), this association attenuated completely to null in the within-family sibling-matched analysis (HR = 0.99, 95% CI [0.74, 1.31]). The apparent long-term risk is attributable to unmeasured genetic, familial, and maternal health indications rather than surgical delivery.

EFFECT SIZE AND PERSISTENCE

The immediate physiological effect is clinically observable: transient tachypnea occurs in roughly 1–3% of planned cesarean births versus less than 1% of vaginal deliveries, but respiratory stabilization typically occurs within 48 to 72 hours under standard neonatal observation. Long-term developmental effect sizes are statistically indistinguishable from zero in genetically and sibling-controlled designs.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: largely confounded (>50%) (population observational associations between cesarean delivery and long-term neurodevelopment attenuate to the null in within-family sibling-comparison designs).

Confounding by indication is severe: elective and emergency cesareans are performed for clinical reasons (e.g., fetal distress, placental insufficiency, maternal hypertension, gestational diabetes, abnormal presentation) that independently associate with child health and development. Studies failing to control for maternal indication or familial shared genes misattribute pre-existing vulnerabilities to the delivery procedure.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean cesarean delivery is a trivial intervention without acute surgical or respiratory considerations. Does not mean delivery mode determines a child's character, cognitive trajectory, or psychiatric future. A cesarean delivery is an obstetric management path, not an intellectual verdict. Does not mean mild newborn jaundice requires home sunlight exposure; hospital phototherapy provides safe, monitored bilirubin clearance when clinical guidelines indicate treatment.

WHAT ACTION IS REASONABLY JUSTIFIED

Obstetric decisions regarding delivery mode should be determined by maternal and fetal medical indications. When pre-labor cesareans are clinically required, neonatal teams should monitor early respiratory transition for transient retained lung fluid. Parents can be reassured that mode of delivery does not harm long-term cognitive or behavioral development.

VERIFIED SOURCES & DOI IDENTIFIERS

- Curran et al. (2015), JAMA Psychiatry (DOI: 10.1001/jamapsychiatry.2015.0846 (https://doi.org/10.1001/jamapsychiatry.2015.0846))
- Jain & Dudell (2006), Seminars in Perinatology (DOI: 10.1053/j.semperi.2006.07.011 (https://doi.org/10.1053/j.semperi.2006.07.011))

Two Languages Do Not Confuse Young Children

Simultaneous dual-language exposure from birth does not confuse infants, impair communicative milestones, or delay core speech discrimination; vocabulary is distributed across languages, and bilingualism does not automatically confer an executive function bonus.

FUSION; MODERATE-INDIVIDUAL-TRAJECTORIES; LOW-AUTOMATIC-BONUS | STATUS: DRAFT | DOMAIN: LANGUAGE, LEARNING | R

WHAT WE KNOW (Narrowest Supported Conclusion)

The human infant brain is natively equipped to acquire more than one language simultaneously without cognitive confusion or communicative delay. Newborns exposed to two distinct languages prenatally demonstrate equal interest in and discrimination between both languages from the first days of life. Across early childhood, dual-language learners reach core linguistic milestones (such as first words, word combinations, and syntactic complexity) on the same timetable as monolinguals when evaluated on total conceptual vocabulary. Code-mixing and language blending are normal signatures of proficient pragmatic communication, not evidence of linguistic pathology.

EVIDENCE AND CAUSAL ASSESSMENT

Byers-Heinlein, Burns & Werker (2010), Psychological Science, tested newborns (mean age: 54 hours) born to monolingual English mothers versus bilingual English-Tagalog mothers using high-amplitude sucking procedures. While English monolingual newborns preferred English over Tagalog, bilingual newborns exhibited equal preference for and robust discrimination between both maternal languages, demonstrating that the perceptual mechanisms for dual-language acquisition are operative from birth.

National Academies of Sciences, Engineering, and Medicine (2017), Promoting the Educational Success of Children and Youth Learning English, conducted a comprehensive consensus synthesis (Conclusion 4-2), confirming that dual-language development does not compromise English proficiency or native-language competence given adequate input. Monolingual single-language standardized assessments routinely misclassify dual-language learners because conceptual vocabulary is distributed across both linguistic environments.

The causal finding is developmental and communicative: the cognitive system accommodates multiple linguistic codes without mutual interference.

EFFECT SIZE AND PERSISTENCE

There is no single effect size for "bilingualism." Single-language vocabulary counts in dual-language toddlers are often smaller than in monolingual peers, but total conceptual vocabulary (concepts named in either language) is equal or superior. Claims of a generalized "bilingual executive function advantage" in young children fail to replicate reliably in large meta-analyses (displaying effect sizes indistinguishable from zero once publication bias and socioeconomic covariates are controlled).

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (developmental psycholinguistic milestone achievement and perceptual discrimination, not an observational population phenotype association).

Severe assessment bias exists in monolingual testing: evaluating dual-language learners exclusively in the societal majority language manufactures artificial "language delay." Confounding by immigrant status, household poverty, and educational policy is frequently misattributed to dual-language exposure itself.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean bilingual children never experience speech or language disorders; clinical speech disorders occur at the same baseline rate in bilinguals as in monolinguals. Does not mean children learn two languages effortlessly without rich social interaction. Does not license advising immigrant parents to abandon their native language at home.

WHAT ACTION IS REASONABLY JUSTIFIED

Parents and caregivers should speak the language in which they are most fluent, emotionally authentic, and comfortable. Clinicians and educators should assess dual-language children across both languages using conceptual scoring, and should never advise families to drop a heritage language under the mistaken belief that it will accelerate majority-language learning.

VERIFIED SOURCES & DOI IDENTIFIERS

- National Academies of Sciences, Engineering, and Medicine (2017), Promoting the Educational Success of Children and Youth Learning English (DOI: 10.17226/24677 (https://doi.org/10.17226/24677))

- Byers-Heinlein, Burns & Werker (2010), Psychological Science (DOI: 10.1177/0956797609360758 (https://doi.org/10.1177/0956797609360758))

Early Allergen Introduction and Baby-Led Weaning

Randomized trials show that introducing peanut protein between 4 and 11 months reduces peanut allergy by ~80% in high-risk infants (LEAP), while modified baby-led weaning and traditional spoon-feeding produce similar growth and choking rates (BLISS).

GRADE: HIGH | STATUS: DRAFT | DOMAIN: FEEDING, HEALTH | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

High-certainty randomized trial evidence (the LEAP trial) shows that introducing age-appropriate peanut protein between 4 and 11 months of age reduces the prevalence of peanut allergy at age five by approximately 81% compared with strict avoidance. Separate randomized evidence on complementary feeding method (the BLISS trial) shows that modified baby-led weaning—where infants self-feed soft finger foods while parents avoid high-risk choking hazards and include iron-rich foods—produces growth trajectories and choking rates comparable to traditional spoon-feeding of purees.

EVIDENCE AND CAUSAL ASSESSMENT

- LEAP RCT (dutoit-2015-leap, N=640 infants aged 4–11 months with severe eczema and/or egg allergy): Infants randomly assigned to consume at least 6 grams of peanut protein per week distributed across three or more meals had a 1.9% prevalence of peanut allergy at 60 months in the skin-prick-negative cohort, compared with 13.7% in the avoidance group (an 86% relative reduction, $p < 0.001$). Across the combined intention-to-treat cohorts, peanut allergy dropped from 17.2% to 3.2% (an 81% relative reduction).

- BLISS RCT (fangupo-2016-bliss, N=206 infants randomized to modified baby-led introduction to solids vs. usual care): No significant differences were observed in the number of choking episodes at 6, 7, 8, 9, or 12 months between groups. Both groups experienced frequent gagging (a normal protective reflex triggered far forward on the tongue) that declined steeply over the second half of the first year.

EFFECT SIZE AND PERSISTENCE

For early peanut introduction, relative risk reduction is ~81% at age 5 (dutoit-2015-leap), and follow-up persistence trials (LEAP-On) demonstrated that tolerance persisted even after 12 months of subsequent peanut cessation. For feeding format (BLW vs. spoon-feeding), differences in BMI z-scores, iron status, and choking incidence are small to null when parents receive basic safety guidance on food texture and iron content.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: robust (<20%) (randomized controlled trials directly assign feeding exposure, removing family-level atopic and socioeconomic confounding).

Observational comparisons of baby-led weaning versus puree feeding are heavily confounded by maternal education, breastfeeding duration, and household income; randomized assignment in BLISS removes this selection bias and shows that feeding philosophy per se is not a major driver of weight or developmental outcomes.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean infants should ever be given whole peanuts or thick globs of peanut butter (which are severe mechanical airway hazards); peanut protein must be thinned into warm water, breastmilk, formula, or fruit/vegetable puree. It also does not mean parents must choose strictly between purees and finger foods—combining spoon-feeding and soft finger foods is developmentally sound.

WHAT ACTION IS REASONABLY JUSTIFIED

Introduce thinned peanut protein and well-cooked egg around 4 to 6 months once the infant shows postural and oral readiness for solids (especially in infants with eczema, after consulting a pediatrician if eczema is severe), and allow infants to explore both mashed/pureed textures and soft, squishable finger foods while strictly avoiding hard, round choking hazards (whole nuts, whole grapes, raw carrot coins, popcorn).

VERIFIED SOURCES & DOI IDENTIFIERS

- Du Toit et al. (2015), New England Journal of Medicine, LEAP randomized trial (dutoit-2015-leap)
- Fangupo et al. (2016), Pediatrics, BLISS randomized trial on choking and gagging (fangupo-2016-bliss)

Early Childcare Quality and Development

NICHD SECCYD cohort shows modest, replicated links between childcare quality and cognitive/language outcomes, with parenting a stronger predictor; cortisol elevations in daycare are small (d ~ 0.15-0.30) and buffered by caregiver sensitivity.

GRADE: MODERATE | STATUS: DRAFT | DOMAIN: PARENTING, ECOLOGY, EARLY-LEARNING | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Higher-quality non-parental childcare (whether center-based daycare, home-based family care, or an individual nanny) in the first three years is modestly and consistently associated with better cognitive, language, and pre-academic outcomes, but parenting quality is a stronger and more consistent predictor of child development than childcare characteristics. Physiological stress studies (including studies comparing daycare vs stay at home mom cortisol levels) document modest afternoon cortisol elevations (d ≈ 0.15--0.30) in center daycare that attenuate with sensitive caregiver interactions and smaller group sizes.

EVIDENCE AND CAUSAL ASSESSMENT

- NICHD Study of Early Child Care and Youth Development (N=1,364), a prospective longitudinal cohort (not randomized), tracked children from birth through adolescence across multiple U.S. sites.

- Higher-quality care (positive language stimulation, caregiver-child interaction) was associated with better language ability at 15/24/36 months, cognitive development at age 2, and school readiness at age 3.

- More hours in center care predicted more externalizing/behavior problems reported by caregivers and teachers, an effect that persisted in attenuated form through age 15.

- Family characteristics and mother-child relationship quality were stronger and more consistent predictors than childcare quality/quantity in the same cohort.

- Vermeer & van IJzendoorn (2006) meta-analyzed cortisol studies in children attending center daycare versus home care, identifying a modest afternoon cortisol increase (d ≈ 0.15--0.30) in center settings relative to staying home with a stay-at-home parent. Follow-up trials confirm this reflects peer stimulation and group regulation rather than neuroendocrine harm, and is buffered by caregiver sensitivity; individual nanny arrangements and high-quality home care show basal cortisol rhythms mirroring home environments. Deciding between center daycare or a nanny is an ecological calibration of peer social complexity versus one-on-one attention, not a safety threshold.

EFFECT SIZE AND PERSISTENCE

Effects are consistently small-to-modest (β ≈ 0.05--0.12). Both the quality-language link and the quantity-behavior-problems link showed some persistence into adolescence (age 15), though at reduced magnitude versus early childhood. Cortisol afternoon elevation resolves upon return to the home environment.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: largely confounded (>50%) (childcare quality associations attenuate when family selection and maternal characteristics are controlled).

This is a large observational cohort, not randomized; families were not randomly assigned to care types or quality levels, so selection (e.g., higher-SES/higher-investment families choosing higher-quality care or choosing between daycare or nanny arrangements) is a plausible confounder not fully addressed by covariate adjustment. No genetically informed replication identified for this literature.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not establish that center daycare harms child attachment or causes neuroendocrine damage. Modest afternoon cortisol shifts do not equal toxic stress, and choosing between daycare, a nanny, or staying at home with a parent is not a moral referendum or a deterministic cognitive fork.

WHAT ACTION IS REASONABLY JUSTIFIED

Institutional/policy support for childcare quality standards (staff-child ratios, caregiver training, peer group sizing) is proportionate given consistent, replicated (if modest) associations; individual-family prescriptions about hours, daycare versus nanny selection, or home-stay mandates are not justified by this evidence alone.

VERIFIED SOURCES & DOI IDENTIFIERS

- NICHD ECCRN SECCYD overview (2006)
- Vandell, Belsky, Burchinal et al., American Psychologist (2006), effect-size synthesis
- Belsky et al. (2007), Child Development, long-term childcare effects (not the age-15 paper)
- Vandell et al. (2010), age-15 SECCYD follow-up
- Vermeer & van IJzendoorn (2006), Early Childhood Research Quarterly, DOI: 10.1016/j.ecresq.2006.07.004 (https://doi.org/10.1016/j.ecresq.2006.07.004)

Motor Development Is Embodied Perception–Action Coupling

Opportunities for infant action depend on the current body, the environment, and social practice; new motor skills change what can be learned, but earlier sitting or walking is not a general cognitive advantage or a parenting score.

GRADE: MODERATE-FRAMEWORK; LOW-MILESTONE-COGNITION | STATUS: DRAFT | DOMAIN: LEARNING | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Infant motor development is not a staircase of postures detached from psychology. Adolph & Hoch (2019) summarize four features: movement is embodied (what the body can do now), embedded (what this environment affords), enculturated (social and cultural practice shapes how and when skills appear), and enabling (a new skill changes what can be explored next). Reaching, crawling, and walking reorganize the information available to the learner.

EVIDENCE AND CAUSAL ASSESSMENT

Review of laboratory and home-observation programs, including work on behavioral flexibility when bodies and environments change. This is not one RCT of “more tummy time causes IQ.” Adjacent: infant-contingency-learning.md is action–outcome learning under a ribbon; this card is the broader body–world coupling.

EFFECT SIZE AND PERSISTENCE

There is no single effect size for “embodiment.” Enabling cascades are task-specific. A new posture can open a problem without guaranteeing a lasting cognitive gain.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Cross-cultural timing differences are easily misread as deficits. Laboratory floors are not crowded apartments.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean infants must be hurried through milestones. A milestone is not a score. Does not mean a delayed walker has a lesser mind. Does not license stimulation products as cognitive engineering.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary opportunities to move in a safe space are consistent with the framework. Milestone ranking and acceleration programs do not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Adolph & Hoch (2019), Annual Review of Psychology (PMC6320716)

Executive Function as Unity and Diversity

Executive function is a family of related but not identical control processes (inhibition, working memory, shifting); early performance is context-bound and is not intelligence, obedience, or character.

-CONSTRUCT; MODERATE-TODDLER-STRUCTURE; LOW-SINGLE-TASK | STATUS: DRAFT | DOMAIN: EXECUTIVE-FUNCTION, COGNITIO

WHAT WE KNOW (Narrowest Supported Conclusion)

Executive function (EF) names top-down processes that help maintain goal-directed behavior when a more automatic response is insufficient. Miyake et al. (2000) showed in adults that inhibition, updating, and shifting share variance and are separable. Garon, Bryson & Smith (2008) mapped an integrative, developmental framework for preschoolers: early forms of attention, inhibition, and working memory are visible before a stable adult factor structure. Toddler performance is a precursor, not miniature adult EF.

EVIDENCE AND CAUSAL ASSESSMENT

The evidence is psychometric, experimental, and developmental, not a causal law of parenting. Task performance includes language, speed, motor demands, motivation, and familiarity. Garon et al. (2008) is a review, not a new RCT. Delay-of-gratification prediction is a different, attenuated claim on the Watts card.

EFFECT SIZE AND PERSISTENCE

Age-related gains across early childhood are robust at the population level. Rank-order stability is lower in toddlerhood than later. A developmental increase is not a permanent individual difference.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a for the construct claim. Task impurity, low reliability, cultural familiarity, and language load are the main threats. SES–EF associations exist in other literatures and are not this card's identification.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

EF is not intelligence, obedience, moral worth, or temperament. High compliance can occur without flexible goal maintenance. Poor task performance can reflect stress, language, or an invalid task. Early differences are not a verdict on children or parents.

WHAT ACTION IS REASONABLY JUSTIFIED

Use multiple observations across contexts. High-stakes ranking of toddlers by one EF task does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Garon, Bryson & Smith (2008), Psychological Bulletin
- Miyake et al. (2000), Cognitive Psychology

Exploration as Sampling: Why Babies Drop Things from High Chairs and Test Physical Rules

Infants drop spoons, toys, and food from high chairs not out of defiance, but as active empirical sampling; surprise from violated physical expectations reorganizes directed exploration and hypothesis testing.

GRADE: MODERATE-LAB; LOW-TRANSFER | STATUS: DRAFT | DOMAIN: LEARNING, MOTOR-COGNITION | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

A universal parenting question is why do babies drop things from high chairs repeatedly and watch them hit the floor. Rather than purposeful naughtiness, oppositional behavior, or random motor clumsiness, cognitive science shows that dropping objects from high chairs is active sensory-motor hypothesis testing. Stahl & Feigenson (2015) in Science tested 11-month-old infants (N = 110) who observed objects either behave predictably or violate core physical principles (solidity or gravitational support). Infants did not merely look longer at the violation; the surprise fundamentally reorganized their subsequent behavior. Crucially, infants directed property-specific test actions toward the object: infants who saw a support violation (an object seemingly suspended in mid-air) repeatedly dropped that object off a surface to test its gravity and support, whereas infants who saw a solidity violation (a ball passing through a hidden wall) repeatedly banged the object against the table to test its solidity.

EVIDENCE AND CAUSAL ASSESSMENT

Stahl & Feigenson (2015) provided clean experimental isolation by contrasting identical objects undergoing expected versus unexpected physical events. The property-specific nature of the exploratory actions (dropping after support violations, banging after solidity violations) rules out general arousal or non-specific excitement. In the high chair, repeated dropping allows the infant to map gravitational trajectory, auditory impact feedback, acoustic resonance, and caregiver social contingency across changing motor grips and trajectories.

EFFECT SIZE AND PERSISTENCE

In controlled laboratory tasks, infants explored unexpected objects significantly longer ($d \approx 0.65$) and demonstrated enhanced statistical word learning for labels paired with surprising objects compared to familiar baseline objects.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. The experimental design uses within-task controls. Household noise, parental fatigue, and temperament individual differences in sensory reactivity moderate how frequently infants repeat physical tests in daily dining environments.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean babies dropping things from the high chair are deliberately trying to infuriate parents or test emotional boundaries. Does not mean parents must pick up dropped items fifty times in a row without setting practical limits. Does not mean high-tech educational toys or physics curricula for babies enhance curiosity beyond everyday household objects.

WHAT ACTION IS REASONABLY JUSTIFIED

Parents can reframe repeated high-chair dropping as natural empirical exploration rather than defiance. Practical management includes offering unbreakable cups and spoons, playing interactive floor-based rolling and dropping games, and calmly removing the spoon or ending the meal when food throwing transitions from curiosity to satiety or fatigue.

VERIFIED SOURCES & DOI IDENTIFIERS

- Stahl, A. E., & Feigenson, L. (2015). Observing the unexpected enhances infants' learning and exploration. Science, 348(6230), 91–94. DOI: 10.1126/science.aaa3799

Fathers, Kin, and Siblings Are Heterogeneous Network Members

Positive father involvement is associated with early learning at small-to-moderate average size; involvement is not one exposure, and fathers, kin, and siblings are not interchangeable doses of care.

GRADE: MODERATE-ASSOCIATION; LOW-UNIQUE-CAUSE | STATUS: DRAFT | DOMAIN: SOCIAL, FAMILY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Fathers, grandparents, and siblings can participate in a child's network through distinct functions—care, play, material support, reciprocity with a less accommodating partner. McWayne, Downer, Campos & Harris (2013) meta-analyzed 21 studies of father involvement and children's early learning: average associations were small to moderate. Quantity of contact is a weaker object than interaction quality. The existing multiple-caregiver card already holds that network size is not a dose.

EVIDENCE AND CAUSAL ASSESSMENT

The McWayne synthesis is observational. Selection (who can be present, who is measured) and correlated family resources are live threats. Dagan & Sagi-Schwartz (2018) treat mother–father networks as configurations, not substitutes. Grandparent and sibling literatures named in C08-P1 remain heterogeneous reviews, not this card's identification. Adjacent: toddler-peer-interaction.md is extrafamilial; this card is intrafamilial network members.

EFFECT SIZE AND PERSISTENCE

Small-to-moderate average associations. They do not establish unique or permanent father effects. Sensitivity and interaction quality are more interpretable than frequency of contact.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: untested-in-GI (father and kin involvement correlates with family socioeconomic resources and unmeasured genetic factors; quantitative GI attenuation estimates absent in this literature).

Samples often overrepresent Western, relatively stable families. Child-to-parent effects (a difficult toddler reducing measured "involvement") are plausible.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Fathers are not mothers' substitutes, and mothers and fathers are not assigned rigidly gendered developmental functions. More caregivers are not automatically better. A missing father-figure style of play is not a developmental sentence. Kin presence is not automatically protective.

WHAT ACTION IS REASONABLY JUSTIFIED

Policies that enable sustained, responsive participation (leave, flexible work, mental-health support) are more defensible than prescriptions about a particular style of father play. Ranking families by network size does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- McWayne, Downer, Campos & Harris (2013), Early Education and Development
- Dagan & Sagi-Schwartz (2018), Child Development Perspectives

Fetal Autonomic Behavior Indexes Present Maturation

Fetal heart-rate variability and state organization become more patterned in the second half of gestation; these measures support claims about present capacity, not reliable individual forecasts of temperament.

GRADE: MODERATE-PRESENT; LOW-FORECAST | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Across the second half of gestation, fetal heart rate typically falls while heart-rate variability and movement–heart-rate coupling increase, consistent with advancing autonomic regulation. These are now-measures of maturation. They are not a temperament test.

EVIDENCE AND CAUSAL ASSESSMENT

DiPietro et al. (2015) synthesize eight longitudinal studies (n = 740 pairs, 24–36 weeks): predictable neuromaturation, within-fetus stability of rank order, a transitional period near 30 weeks, and deviations under growth restriction. DiPietro's earlier 2007 follow-up found statistical prediction from late-gestation HRV to toddler scores; that is prediction, not a persistent fetal trait, and is modest. Pollatou et al. (2024) link fetal HR/HRV to newborn brainstem/hypothalamic/dACC connectivity in a small imaging sample (n = 60)—a physiologic association, not a character forecast.

EFFECT SIZE AND PERSISTENCE

Primary outcomes are contemporaneous. Later associations, where present, are small-to-moderate and sit inside postnatal development.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (fetal cardiac electrophysiology, not an observational population phenotype association).

Indirect measurement; fetal state and maternal movement; publication bias toward striking findings; IUGR as a clinical, not typical, contrast.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean a "calm" or "active" fetus has a fixed personality, or that fetal HRV is a prenatal IQ score.

WHAT ACTION IS REASONABLY JUSTIFIED

Clinical fetal monitoring in high-risk pregnancy is justified. Using fetal behavior as a parenting forecast is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- DiPietro et al. (2015), Monographs of the Society for Research in Child Development
- Pollatou et al. (2024), Journal of Neuroscience

Fetal Growth Restriction Is a Clinical Outcome, Not a Destiny

INTERGROWTH-21st provides international fetal-size standards; severe restriction is clinically meaningful, but birth size is not a proxy for intelligence, worth, or future personality.

ADE: MODERATE-STANDARDS; HIGH-PERCENTILE-NOT-DESTINY | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFI

WHAT WE KNOW (Narrowest Supported Conclusion)

Fetal growth reflects genes, placenta, maternal condition, and environment. INTERGROWTH-21st constructed international ultrasound standards from low-risk pregnancies so that size can be interpreted against a common chart. Severe restriction is clinically meaningful. Birth size is not intelligence, worth, or personality. Prenatal differences can attenuate, persist, or change form after birth; persistence must be measured (CH02-C13).

EVIDENCE AND CAUSAL ASSESSMENT

Papageorghiou et al. (2014), INTERGROWTH-21st Fetal Growth Longitudinal Study: 4,321 low-risk pregnancies across eight countries; serial ultrasound from 14 weeks; centiles for head circumference, BPD, OFD, abdominal circumference, femur length. This is a standards paper, not a trial of smallness. Continuity after birth is a longitudinal claim: attenuation, persistence, and transformation are all possible; none can be assumed from size alone.

EFFECT SIZE AND PERSISTENCE

Distinguish percentile difference, functional difference, and impairment. Dating error and constitutional smallness are not placental failure.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (normative anthropometric measurement standard, not an observational exposure).

Low-risk INTERGROWTH sample is prescriptive, not a census of all fetuses; collider/selection if only small babies are followed; postnatal nutrition.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean a small newborn is a lesser child, or that catch-up proves the prenatal period was irrelevant.

WHAT ACTION IS REASONABLY JUSTIFIED

Clinical surveillance of true growth restriction is justified. Treating a percentile as a forecast of character is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Papageorghiou et al. (2014), Lancet (INTERGROWTH-21st)

Guided Play Efficacy and Pedagogical Scaffolding

Adult-guided play shows targeted advantages over direct instruction for early mathematics and shape concepts, and over free play for spatial vocabulary; evidence does not show broad benefits for literacy, general executive function, or mandatory home instruction.

RATE-MATH-SPATIAL; LOW-LITERACY-AND-HOME-CURRICULUM | STATUS: DRAFT | DOMAIN: PLAY, EARLY-LEARNING | REPLICAT

WHAT WE KNOW (Narrowest Supported Conclusion)

Play is not a unitary exposure. Guided play sits between child-led free play and adult-directed didactic instruction: adults establish a prepared environment with intentional learning goals and gently scaffold the child's self-directed inquiry without taking over agency. In early educational settings, guided play shows specific, targeted advantages for early mathematical thinking (e.g., shape knowledge, numeracy) compared to didactic teaching, and for spatial vocabulary compared to unguided free play.

EVIDENCE AND CAUSAL ASSESSMENT

Skene et al. (2022), Child Development (systematic review of 39 studies, meta-analysis of 17 studies across 3,893 children), evaluated the effectiveness of guided play relative to direct instruction and free play. Guided play yielded significant positive effects for early mathematics ($g = 0.24$, 95% CI [0.12, 0.35], $n = 1,165$) and shape knowledge ($g = 0.63$) over direct instruction, and for spatial vocabulary ($g = 0.93$, 95% CI [0.10, 1.75], $n = 137$) over free play.

However, the meta-analysis found no statistically reliable advantage for early literacy ($g = 0.28$, 95% CI [−0.22, 0.79], $n = 233$), and no effect on core behavioral regulation ($g = −0.03$, $n = 1,413$) or inhibitory control ($g = −0.06$, $n = 145$) compared to direct instruction. Task switching showed a modest positive effect ($g = 0.40$, $n = 132$).

Weisberg et al. (2016), Current Directions in Psychological Science, delineates the underlying mechanism: guided play harnesses active, engaged, meaningful, and socially interactive learning ("playful learning") while maintaining child autonomy, which constrains the search space for conceptual exploration without triggering the passive disengagement common in drill-and-practice instruction.

EFFECT SIZE AND PERSISTENCE

The observed effect sizes are small-to-moderate for domain-specific preschool numeracy and shape learning ($g = 0.24$ to 0.63). Effects are documented immediately post-intervention in preschool or laboratory contexts; long-term longitudinal persistence into middle childhood and general academic achievement has not been established by randomized trials.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (pedagogical intervention / educational RCT trials, not population observational phenotype).

Substantial heterogeneity exists across studies in how "guided play" is operationalized, ranging from subtly prepared discovery toys to tightly scripted adult prompts. Studies with smaller samples ($n < 150$) reported the largest spatial vocabulary effects. Publication bias in early educational intervention literature remains an acknowledged risk.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean unstructured free play is worthless or should be replaced by structured educational games. Does not mean parents must turn every domestic play session into a stealth pedagogical lesson. Does not mean guided play accelerates overall general intelligence or executive functioning.

WHAT ACTION IS REASONABLY JUSTIFIED

Early childhood educators can deliberately employ guided play to teach complex spatial, mathematical, and geometric concepts more effectively than didactic worksheets. Parents can follow their child's lead, ask open-ended questions, and offer gentle scaffolding when invited, but should feel no obligation to manage or direct ordinary exploratory play.

VERIFIED SOURCES & DOI IDENTIFIERS

- Skene et al. (2022), Child Development (DOI: 10.1111/cdev.13730 (https://doi.org/10.1111/cdev.13730))
- Weisberg et al. (2016), Current Directions in Psychological Science (DOI: 10.1177/0963721416645512 (https://doi.org/10.1177/0963721416645512))

Household Food Insecurity Is Associated with Infant and Toddler Health, Not Parental Failure

In a low-income US clinical sample, household food insecurity is associated with poorer infant and toddler health; the association is not a parenting residual or a cognitive-intervention law.

RADE: MODERATE-ASSOCIATION; LOW-COGNITION-LAW | STATUS: DRAFT | DOMAIN: ECOLOGY, HEALTH | REPLICATION: VERIFIE

WHAT WE KNOW (Narrowest Supported Conclusion)

Cook and colleagues (2004) found that household food insecurity was associated with poorer health and more hospitalizations among infants and toddlers in a low-income US clinical sample, after some covariates. Hunger in the household is a material condition, not a motivation problem.

EVIDENCE AND CAUSAL ASSESSMENT

Observational clinical sample, not an RCT of food. Cross-sectional and short-term health reports can mix selection, reporting, and material deprivation. Adjacent: income-cash-transfer-early-development.md is cash, not food. Prenatal-stress GI attenuation already warns against treating family correlations as parenting effects (prenatal-maternal-stress-anxiety.md). Sleep and attachment-intervention theory are not restated.

EFFECT SIZE AND PERSISTENCE

Odds of fair/poor health and hospitalization were elevated with food insecurity in this sample. Persistence as a developmental cognitive effect is not this identification.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: untested-in-GI (food precarity co-occurs with structural economic stress and parental vulnerabilities; twin/sibling designs not applied to this specific pediatric association).

Low-income clinical samples are not all households. Caregiver report of child health and food can share method variance. Access to care, housing, and income remain entangled.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Food insecurity is not proof of low parental commitment. A food-secure household is not a cognitive guarantee. Poverty's cash-transfer nulls (Baby's First Years age four) do not make food irrelevant.

WHAT ACTION IS REASONABLY JUSTIFIED

Treat food access as a material and institutional problem. Ranking parents by a hunger screen does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Cook et al. (2004), Journal of Nutrition

Income Support Is Not a Guaranteed Cognitive Intervention

Poverty is associated with developmental disadvantage through multiple pathways; Baby's First Years shows a defined monthly cash transfer can change infant brain activity without producing preregistered age-four cognitive gains.

: MODERATE-GRADIENT; LOW-TO-MODERATE-CASH-COGNITION | STATUS: DRAFT | DOMAIN: ECOLOGY, POLICY | REPLICATION: VE

WHAT WE KNOW (Narrowest Supported Conclusion)

Low income is consistently associated with developmental and health disadvantage. That association mixes material deprivation, stress, housing, nutrition, service access, and selection. Baby's First Years randomized mothers to a larger versus smaller monthly cash transfer. Troller-Renfree et al. (2022) reported greater mid-to-high-frequency infant EEG power in the high-cash group (standardized effects about 0.17–0.26). Noble et al. (2025) reported no statistically significant differences on preregistered language, executive-function, social-emotional, and related developmental outcomes at age four.

EVIDENCE AND CAUSAL ASSESSMENT

BFY is an RCT of a defined transfer, not an estimate of eliminating poverty. Observational gradients are not this card's causal claim. Adjacent GI-attenuated cards (prenatal stress; breastfeeding IQ) already warn against treating family-level associations as parenting effects. Childcare quality remains a different exposure.

EFFECT SIZE AND PERSISTENCE

Report the infant EEG effect separately from the age-four nulls. Do not infer that a small early neural difference is a durable developmental benefit, or that a null result means poverty is harmless.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (randomized controlled cash transfer trial, not an observational parenting phenotype).

Effects may depend on transfer size, duration, baseline deprivation, local prices, and other services. The trial does not estimate broader anti-poverty packages.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Poverty is not evidence of low parental commitment. The BFY null does not show that material conditions have no causal effects. Cash is not a substitute for housing, health care, or childcare quality.

WHAT ACTION IS REASONABLY JUSTIFIED

Treat income support as a material-security and anti-poverty policy first. Evaluate developmental outcomes directly rather than promising universal cognitive gains.

VERIFIED SOURCES & DOI IDENTIFIERS

- Troller-Renfree et al. (2022), PNAS
- Noble et al. (2025), NBER working paper w33844

Infant Attention Is a Family of Processes, Not a Trait

Attention in infancy comprises dissociable functions (alertness, orienting, feature attention, endogenous control) that change on different schedules; inattention in a given hour is not a character verdict.

GRADE: MODERATE-FUNCTIONS; LOW-TRAIT | STATUS: DRAFT | DOMAIN: COGNITION | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Infants do not process every available signal equally. Attention selects. Colombo (2001) treats infant visual attention as several functions—alertness, spatial orienting, attention to object features, and endogenous/voluntary control—that are present in rudimentary form early and then change on dissociable postnatal schedules.

EVIDENCE AND CAUSAL ASSESSMENT

Colombo (2001), Annual Review of Psychology, is a research synthesis of looking, habituation, and related work, not an RCT of "attention training." The review's claim is mechanistic and taxonomic: different attentional systems mature at different rates, so the role of attention in development is an interaction among systems at different levels of maturity. Everyday causal effects of changing attention on later learning are harder to isolate than laboratory associations.

EFFECT SIZE AND PERSISTENCE

No portable effect size for "better attention" as a developmental intervention. Look duration and habituation rate have been used as individual-difference measures; they are not destiny.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (attentional function review, not an observational population phenotype association).

Arousal, hunger, fatigue, distress, overstimulation, and posture alter looking. Parent-report "inattentive" and laboratory look duration are not the same construct. Prediction from early looking to later scores is not determination.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean a tired or distressed infant lacks capacity. Does not mean more stimulation produces more attention. Does not mean attention is a parenting score.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary conditions in which an infant can settle, look, and disengage are consistent with the finding. Attention-training products and performance dashboards are not justified here.

VERIFIED SOURCES & DOI IDENTIFIERS

- Colombo (2001), Annual Review of Psychology (DOI 10.1146/annurev.psych.52.1.337)

Babbling Is Exploration, Not an IQ Test

Canonical babbling is motor, auditory, and social practice; late onset can flag later production risk in screened samples and is not a verdict on typical intelligence or parenting.

GRADE: HIGH-CANONICAL-STAGE; LOW-IQ-VERDICT | STATUS: DRAFT | DOMAIN: LANGUAGE | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Canonical babbling — well-formed consonant–vowel syllables, often reduplicated — is vocal exploration shaped by maturation, auditory feedback, and social response. It is not yet speech and not a score of intelligence. Oller, Eilers, Neal & Schwartz (1999): nearly all infants with normal hearing reach the canonical stage by 10 months; among at-risk infants screened then, late onset predicted smaller production vocabularies at 18–30 months. Petitto & Marentette (1991) described rhythmic manual activity in sign-exposed infants: language production is not confined to the vocal tract.

EVIDENCE AND CAUSAL ASSESSMENT

Oller et al. (1999) combine the canonical-stage description with screening of more than 3,400 at-risk infants near 10 months. Late onset predicted later expressive vocabulary in that group. Identification is observational prediction in an at-risk sample, not an RCT of "more babbling." Petitto & Marentette (1991) is a small observational report of manual babbling — a modality boundary, not a spoken-IQ finding.

EFFECT SIZE AND PERSISTENCE

The screening result is a group difference in later production vocabulary. It is not a portable effect size for stimulating babble in typical homes.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (observational clinical screening of vocal milestones, not an observational parenting exposure).

Hearing impairment, motor difference, and illness can delay canonical babbling; those pathways are not typical-home intelligence. Parent report is usable for screening, not acoustic verification. Genetically informed attenuation: n/a.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean a quiet month of vocal play is an intelligence verdict. Does not mean parents must drill syllables. Does not mean spoken babble is the only language production system.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary vocal play and answering a baby's sounds are consistent with a feedback-rich pathway. Using babbling age as a character or IQ score is not justified. Late onset on an at-risk screen is a reason to look at hearing and later production — not a completed diagnosis.

VERIFIED SOURCES & DOI IDENTIFIERS

- Oller, Eilers, Neal & Schwartz (1999), Journal of Communication Disorders (DOI 10.1016/s0021-9924(99)00013-1; PMID 10466095)
- Petitto & Marentette (1991), Science (DOI 10.1126/science.2006424)

Infant–Caregiver Co-Regulation Is Bidirectional

Co-regulation is a bidirectional, context-sensitive process in which infant and caregiver organize arousal together; synchrony is not uniformly beneficial, and nurturing care is an ecology, not a maternal purity test.

ODERATE-EXISTENCE; LOW-TO-MODERATE-MECHANISM | STATUS: DRAFT | DOMAIN: REGULATION, ATTACHMENT | REPLICATION:

WHAT WE KNOW (Narrowest Supported Conclusion)

Human infants have limited capacity to stabilize arousal, temperature, feeding, sleep, and distress alone. Caregivers help organize those states through contingent, repeated interaction. The process is bidirectional: infant signals shape the adult; adult response shapes the infant. WHO/UNICEF/World Bank Nurturing Care treats responsive caregiving as one component of an ecology that also includes health, nutrition, safety, and early learning.

EVIDENCE AND CAUSAL ASSESSMENT

Feldman (2012) models affiliative bonds as time-locked coordination of physiology and interactive behavior; it is a theoretical synthesis of observational and psychophysiological work, not an RCT of "synchrony training." The Nurturing Care framework is a policy synthesis, not a causal estimate of ordinary parenting differences. Existing cards already cover sensitivity-training RCTs (responsive-caregiving-attachment-interventions.md) and sleep-training RCTs (infant-sleep-training.md); those claims are not restated here.

EFFECT SIZE AND PERSISTENCE

There is no single effect size for "co-regulation." Immediate changes in arousal during an interaction are more defensible than permanent biological programming. Long-run executive-function or personality claims require separate identification.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (bidirectional dynamic interaction framework, not an observational dosage law).

Infant distress can move caregiver physiology (child-to-parent effects). Matching can be engagement, recovery, or stress contagion depending on timing and system. Material conditions (leave, housing, health services) enable or constrain responsiveness; treating the parent as a sole exposure is a specification error.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Responding to crying is not automatically "spoiling," and waiting is not automatically "teaching independence." Synchrony is not a purity test. Nurturing care is not a demand that one adult regulate every state perfectly.

WHAT ACTION IS REASONABLY JUSTIFIED

Support contingent, developmentally appropriate interaction and the conditions that make it possible. Promises of biological optimization, or using this card to rewrite the sleep and attachment cards, are not justified.

VERIFIED SOURCES & DOI IDENTIFIERS

- WHO, UNICEF & World Bank Group (2018), Nurturing care for early childhood development
- Feldman (2012), Parenting: Science and Practice

Infant Contingency Learning

Infants can learn that a self-generated action predicts a sensory consequence in the mobile-conjugate-reinforcement paradigm; this is action–outcome learning under controlled conditions, not adult-like causal theory or a stimulation mandate.

GRADE: HIGH-CONTROLLED; LOW-TRANSFER | STATUS: STABLE | DOMAIN: LEARNING, PLAY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

In the mobile-conjugate-reinforcement paradigm, a ribbon links an infant's ankle to an overhead mobile. Kicking moves the mobile; later kicking above baseline indexes retention of that action–outcome relation. The design manipulates whether the infant's own action produces the consequence, which is stronger identification than passive observation.

EVIDENCE AND CAUSAL ASSESSMENT

Rovee & Rovee (1969) established conjugate reinforcement of infant kicking. Subsequent work, summarized in methods papers (e.g. Merz et al. 2017) and a 2025 review of the last quarter-century, treats the paradigm as a reliable laboratory measure of learning and memory: speed of acquisition and length of retention increase with age; reminders can reactivate a forgotten response, often with high context specificity. Incomplete processing parameters and the later turn toward individual differences and agency do not block the narrow High claim that the contingency can be acquired and retained under matching conditions — the same split as Saffran laboratory segmentation versus household transfer.

EFFECT SIZE AND PERSISTENCE

There is no single standardized effect for "infant learning." The defensible unit is behavioral acquisition and a retention window that varies with age, training distribution, and contextual match. A weak test response can be retrieval or motor-performance failure rather than absent learning.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (laboratory operant conditioning / infant contingency reinforcement, not an observational population phenotype association).

Arousal, interest in the mobile, motor capacity, and the opportunity to kick all affect the score. Laboratory contingency is not everyday exploration. Agency (whether the infant experiences the relation as self-caused) remains an interpretive question.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean infants are miniature scientists, that every repeated action is conscious goal pursuit, or that parents must run training protocols. Early task performance is not destiny.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary opportunities to act and notice consequences are consistent with the finding. Intensive stimulation programs and performance monitoring do not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Rovee & Rovee (1969), Journal of Experimental Child Psychology
- Merz et al. (2017), mobile-paradigm methods (PMC5716629)
- Cuevas et al. (2025), Infant Behavior & Development (PMC12354038)

Infant Crying Is a Graded Signal, Not a Code

Crying recruits care and conveys degree of distress; it is not a reliable one-to-one code for hunger, pain, or parental error, and ordinary inability to 'decode' the sound is not failure.

RADE: MODERATE-SIGNAL; LOW-ACOUSTIC-DECODING | STATUS: DRAFT | DOMAIN: BIRTH, REGULATION | REPLICATION: VERIFIE

WHAT WE KNOW (Narrowest Supported Conclusion)

Infant crying is an adaptive distress signal that usually elicits caregiving. It is a graded signal of how upset the infant is, not a catalog of distinct "hunger," "pain," and "boredom" cries that a competent adult can decode from acoustics alone. Context, not timbre, usually tells the caregiver what to try.

EVIDENCE AND CAUSAL ASSESSMENT

Zeifman & St James-Roberts (2017) review parenting-the-crying-infant research and note that distinct cry-type decoding has been largely refuted; Gustafson, Wood & Green's review in Barr, Hopkins & Green (2000) is the cited careful treatment. The 2023 scientometric map of cry/caregiver papers (PMC10297674) still treats Barr's 1990 "normal crying curve" as foundational. Identification is review-level and observational, not an RCT of "decoding training."

EFFECT SIZE AND PERSISTENCE

No universal effect size. Crying typically rises over the first weeks and falls by a few months in many infants; that population curve is not a parenting-quality score. Persistence of "colic" is a different clinical claim.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (developmental acoustic physiology, not an observational population phenotype association).

Parent report vs recording; prematurity, illness, feeding, and temperament; reverse causation if caregivers under strain hear more crying. Studies that treat duration as a child or parent trait risk blame.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Inability to name the cause from the sound is not parental failure. Prolonged crying is not automatically proof of harm or of spoiling. Crying is not a language with a dictionary.

WHAT ACTION IS REASONABLY JUSTIFIED

Realistic expectations, support, and attention to danger signs are justified. Training parents to hear "types" of cry as a skill test is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Zeifman & St James-Roberts (2017), Current Opinion in Psychology (PMC5494986)

Infants Can Form Attachments to More Than One Caregiver

Infants can organize relationship-specific attachment patterns with more than one caregiver; network size is not a dose, and a second caregiver is not automatically better.

ADE: MODERATE-OCCURRENCE; LOW-NETWORK-DOSE | STATUS: DRAFT | DOMAIN: ATTACHMENT, SOCIAL | REPLICATION: VERIFI

WHAT WE KNOW (Narrowest Supported Conclusion)

Most children are raised by more than one caregiver and can form attachment bonds with more than one of them. Patterns toward mothers and fathers can be concordant or discordant. Dagan & Sagi-Schwartz (2018, 2021) argue that the scientifically useful object is a network configuration, not a single caregiver-child score treated as the child's relational world.

EVIDENCE AND CAUSAL ASSESSMENT

The literature is primarily observational and longitudinal. Sensitivity is related to security across caregivers, but caregiver sensitivity is also shaped by child behavior, stress, resources, and history. Studies are few, often small, and historically mother-centric. This is not a randomized trial of "more adults."

EFFECT SIZE AND PERSISTENCE

There is no agreed dose-response for number of caregivers. Persistence is relationship-specific and revisable. Early classifications do not determine adult functioning on this card.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a for the occurrence claim. Samples often overrepresent Western, relatively stable two-parent households. Professional caregivers, kin, and fathers are unevenly measured.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean children need a single exclusive caregiver. Does not mean more caregivers are better. Does not mean group care, or a nanny, is interchangeable with a parent. Does not license using BEIP deprivation effect sizes (severe-institutional-deprivation.md) as a description of ordinary extrafamilial care.

WHAT ACTION IS REASONABLY JUSTIFIED

Support stable access to responsive caregivers and reduce avoidable disruptions. Do not prescribe one universal family structure.

VERIFIED SOURCES & DOI IDENTIFIERS

- Dagan & Sagi-Schwartz (2021), New Directions for Child and Adolescent Development (PMID 34936190)
- Dagan & Sagi-Schwartz (2018), Child Development Perspectives

Infant Perceptual Narrowing for Speech Contrasts

During the first year, discrimination of native-language phonetic contrasts is maintained or strengthened while sensitivity to some nonnative contrasts declines; this is attunement, not a permanent closing of language learning.

ADE: HIGH-LAB; MODERATE-TIMETABLE; LOW-SEALED-DEADLINE | STATUS: STABLE | DOMAIN: LANGUAGE | REPLICATION: VERIF

WHAT WE KNOW (Narrowest Supported Conclusion)

Young infants can discriminate phonetic contrasts they have not yet needed for a native language. Across the first year, that broad sensitivity is reorganized by experience: native-language contrasts remain discriminable, while some nonnative contrasts become harder. This is attunement to the languages actually heard, not a deletion of a universal dictionary.

EVIDENCE AND CAUSAL ASSESSMENT

Werker & Tees (1984) compared English infants with English and Salish adults on a Salish (non-English) contrast, and combined cross-sectional with longitudinal infant data. Infants discriminated the nonnative contrast; the decline appeared within the first year and depended on specific language experience. Kuhl, Williams, Lacerda, Stevens & Lindblom (1992), Science, showed in an independent laboratory that by six months American infants' vowel perception had already been altered by English experience — a conceptual replication of experience-dependent phonetic attunement, not a household-vocabulary study and not a universal 6–12-month clock. Identification is experimental (manipulated contrast × age/experience), not a randomized life-course trial. A 2002 reprint of Werker & Tees remains a landmark; later work has questioned how far the original samples generalize. The timetable and sealed-deadline claims are not replicated as universals and remain Moderate and Low.

EFFECT SIZE AND PERSISTENCE

The outcome is contemporaneous discrimination, not later vocabulary. Persistence of the decline into adulthood is well known for some contrasts; recovery and bilingual maintenance show that the system is not sealed.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (laboratory phonetic discrimination task, not an observational population phenotype association).

Historically narrow language samples; looking-time methods; infant state. Diversity of languages and communities remains a live critique of treating 6–12 months as a single clock.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean a rigid deadline after which a second language cannot be learned, or that hearing more than one language confuses the infant. Prenatal preference (DeCasper) is a different claim.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary participation in a linguistic community is the relevant exposure. Phonetic-discrimination training as cognitive enhancement is not justified by this card.

VERIFIED SOURCES & DOI IDENTIFIERS

- Werker & Tees (1984), Infant Behavior and Development
- Kuhl et al. (1992), Science 255:606–608

Early Physical Reasoning and Object Permanence: Looking vs. Manual Search

Young infants demonstrate object permanence and physical reasoning in looking-time tasks months before manual cloth retrieval; search failure reflects frontostriatal inhibitory and motor planning immaturity rather than absent representation.

GRADE: MODERATE-VOE; LOW-COMPLETE-THEORY | STATUS: DRAFT | DOMAIN: LEARNING, COGNITION | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Parents frequently ask when do babies get object permanence? For half a century, the Piagetian orthodoxy held that infants do not acquire object permanence until approximately 8 to 9 months, when they first reach under a cloth to retrieve a hidden rattle, and that full mental representation remains incomplete until 18 to 24 months as evidenced by perseverative A-not-B reach errors (Piaget 1954). However, modern looking-time experiments demonstrated that young infants possess mental representations of hidden objects far earlier. Baillargeon (1987) demonstrated using rotating-drawbridge habituation displays that 3.5- to 4.5-month-old infants look significantly longer at impossible occlusion events (where a solid screen rotates 180 degrees through the space occupied by a hidden wooden box), proving that perceptual awareness of object solidity and permanence exists months before infants coordinate reaching hands. Diamond (1985) subsequently showed that the A-not-B search error does not stem from absent object permanence, but from the slow maturation of dorsolateral prefrontal cortex inhibitory control and motor working memory: infants look toward the correct hiding location B while their reaching hand perseverates to previously reinforced location A.

EVIDENCE AND CAUSAL ASSESSMENT

Experimental dissociation between looking and manual reaching. Baillargeon's violation-of-expectation (VOE) paradigm has been replicated across occlusion, containment, collision, and support configurations (Stahl & Feigenson 2015). Diamond's developmental progression across 7.5 to 12 months showed that increasing the delay between hiding and reaching produces systematic A-not-B perseveration until prefrontal inhibitory circuits mature. Search tasks and looking tasks dissociate because two-step means-end manual retrieval places heavy demands on immature motor executive coordination.

EFFECT SIZE AND PERSISTENCE

VOE looking-time differences between possible and impossible occlusion events are robust (d = 0.50 to 0.80 in tightly controlled laboratory displays). Longitudinal tracking shows that early VOE sensitivity reflects normative cognitive architecture rather than an individual predictor of school-age IQ.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Methodological debates center on whether prolonged staring reflects rich conceptual representation versus low-level perceptual novelty or habituation dynamics (Aslin 2007; ManyBabies 2020).

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean young babies are miniature physicists who understand abstract gravity or mechanics. Does not mean failing to reach for a hidden toy at 6 months indicates cognitive delay. Does not mean commercial flashcards or baby-physics toys accelerate brain development.

WHAT ACTION IS REASONABLY JUSTIFIED

Parents do not need special cognitive drilling toys or apps. Natural interactive games (peekaboo, hiding a toy under a blanket, watching objects roll and fall) provide ample sensory-motor scaffolding matching the infant's natural developmental timetable.

VERIFIED SOURCES & DOI IDENTIFIERS

- Baillargeon, R. (1987). Object permanence in 3.5- and 4.5-month-old infants. Developmental Psychology, 23(5), 655–664. DOI: 10.1037/0012-1649.23.5.655

- Diamond, A. (1985). Development of the ability to use recall to guide action, as indicated by infants' performance on AB. Child Development, 56(4), 868–883. DOI: 10.2307/1130099

- Stahl, A. E., & Feigenson, L. (2015). Observing the unexpected enhances infants' learning and exploration. Science, 348(6230), 91–94. DOI: 10.1126/science.aaa3799

Pointing Is Shared Reference, Not a Language Product

Declarative pointing is concurrently and longitudinally associated with language; commercial baby-sign packages are a different, inconclusive claim.

GRADE: MODERATE-REFERENCE; LOW-PRODUCT | STATUS: DRAFT | DOMAIN: LANGUAGE, SOCIAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Pointing can coordinate two minds around one referent. Colonnesi, Stams, Koster & Noom (2010) found pointing associated with language concurrently (r = .52) and longitudinally (r = .35) across 25 studies (734 children), for declarative/general pointing, not imperative pointing. Commercial baby-sign packages for typically developing hearing infants are a different claim: Fitzpatrick et al. (2014) found inconclusive language advantages and no evidence of harm.

EVIDENCE AND CAUSAL ASSESSMENT

Colonnesi et al. (2010), Developmental Review, is an observational meta-analysis. Shared attention, motor, and learning capacities can drive both gesture and speech. Longitudinal r = .35 is prediction, not an RCT of "more pointing." Fitzpatrick, Thibert, Grandpierre & Johnston (2014) reviewed gestural-communication programs in typically developing hearing infants under 36 months: language benefits inconclusive; one randomized study found no advantage on standardized measures. Joint attention (Mundy & Newell) is not re-extracted here.

EFFECT SIZE AND PERSISTENCE

Concurrent association is large by social-development standards; longitudinal association is moderate. Neither is a treatment effect for a marketed curriculum.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (developmental communicative milestone, not an observational population phenotype association).

Age, pointing motive, and language modality moderate the Colonnesi estimates. SES and country were examined as moderators; they do not license a parental purity test. Baby-sign trials are small and mixed. Genetically informed attenuation: n/a.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean a quiet pointer is delayed. Does not mean baby-sign products are required, harmful, or equivalent to naturally emerging gesture. Does not mean Hart & Risley.

WHAT ACTION IS REASONABLY JUSTIFIED

Treat pointing and showing as ordinary shared reference. Purchasing a sign package to accelerate typical speech is not justified here.

VERIFIED SOURCES & DOI IDENTIFIERS

- Colonnesi, Stams, Koster & Noom (2010), Developmental Review (DOI 10.1016/j.dr.2010.10.001)
- Fitzpatrick, Thibert, Grandpierre & Johnston (2014), First Language (DOI 10.1177/0142723714562864)

Infant Sleep Training: Graduated Extinction & The Ferber Method

RCT and cohort evidence finds no measured attachment or behavioral harm from behavioral sleep training, graduated extinction (the Ferber method), and cry-it-out through age 6.

GRADE: LOW-TO-MODERATE | STATUS: DRAFT | DOMAIN: REGULATION, HEALTH, PARENTING | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Randomized and longitudinal evidence does not find that graduated extinction (the Ferber method), unmodified extinction (Marc Weissbluth / cry it out / CIO), gentle sleep coaching, bedtime fading, or related behavioral infant sleep training (applied across nighttime sleep, daytime naps, crib transitions, and pacifier routines, including commercial schedules such as Moms on Call, Taking Cara Babies, Precious Little Sleep, Babywise / Bucknam & Ezzo, Suzy Giordano, Harvey Karp, Craig Canapari, Tracy Hogg, Gina Ford, Penelope Leach, and James McKenna) harms measured infant–parent attachment or increases behavioral/emotional problems through at least age 6, though most trials are underpowered for rare or long-latency harms.

EVIDENCE AND CAUSAL ASSESSMENT

- Price, Wake, Ukoumunne & Hiscock (2012), Pediatrics: five-year follow-up of a randomized infant sleep intervention found no differences in child sleep, mental health, or child–parent relationship at age 6. The previously listed Sleep Health (2017) paper by those four authors was not located.

- Gradisar et al. (2016), Pediatrics RCT (n=43): graduated extinction and bedtime fading improved some sleep measures versus education control. At 12 months there were no significant group differences in Strange Situation attachment or emotional/behavioral problems; salivary cortisol did not show an adverse stress signal. This is the 2016 Pediatrics paper. It is not a 52-study meta-analysis.

- Bilgin & Wolke (2020), JCPP: observational cohort of 178 dyads. Maternal report of leaving the infant to cry it out in the first six months was not associated with insecure or disorganized attachment at 18 months. This is not an RCT of a sleep-training protocol.

- Mindell, Kuhn, Lewin, Meltzer & Sadeh (2006), SLEEP: AASM review of 52 treatment studies (n>2,500), nine of them Level I RCTs. Behavioral interventions were generally reported as efficacious for bedtime problems. This is an efficacy synthesis, not a 2016 Pediatrics harm meta-analysis. The retired catalog ID hiscock-2016 identified none of these papers.

- Williamson, Mindell, Hiscock & Quach (2019) describes LSAC sleep-problem trajectories. It is not a sleep-training RCT and does not license claims about extinction methods, cortisol, or attachment.

EFFECT SIZE AND PERSISTENCE

Reported effects are null for attachment security and behavior problems in the RCTs and follow-ups located here, tracked out to roughly age 6 in the longest follow-up. Sleep improvements in Gradisar 2016 and in Mindell 2006's efficacy review are a different claim from harm.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (randomized behavioral intervention trials and follow-ups, not an observational phenotype).

Most trials are relatively small to moderate and rely heavily on parent-report sleep measures; Gradisar's attachment comparison is underpowered. Self-selection into sleep-training studies may limit generalizability. Bilgin & Wolke is observational. Longer-term outcomes and effects in infants with heightened biological vulnerability are not well covered.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not establish that sleep training is necessary or that not sleep-training risks poorer outcomes; absence of demonstrated harm is not equivalent to a demonstrated universal safety guarantee at the individual level.

WHAT ACTION IS REASONABLY JUSTIFIED

Families choosing structured sleep-training methods are not acting against the weight of current evidence on attachment or behavioral harm; this is a values/preference-consistent decision domain more than an evidence-mandated one.

VERIFIED SOURCES & DOI IDENTIFIERS

- Price, Wake, Ukoumunne & Hiscock (2012), Pediatrics, five-year follow-up of the Melbourne infant sleep RCT

- Gradisar et al. (2016), Pediatrics RCT (DOI 10.1542/peds.2015-1486)
- Bilgin & Wolke (2020), JCPP (DOI 10.1111/jcpp.13223)
- Mindell et al. (2006), SLEEP AASM review (DOI 10.1093/sleep/29.10.1263)
- Williamson, Mindell, Hiscock & Quach (2019), LSAC sleep-problem trajectories (not a sleep-training RCT)

Infant Social Referencing Under Uncertainty

Late in the first year, infants can use a familiar adult's emotional signal to regulate approach or withdrawal in an ambiguous situation; referencing is not an attachment classification.

GRADE: HIGH-LAB; LOW-ATTACHMENT-VERDICT | STATUS: STABLE | DOMAIN: SOCIAL, REGULATION | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

When a situation is ambiguous, infants toward the end of the first year often look to a familiar adult's face and then approach or withdraw in line with that signal. On the visual cliff, Sorce, Emde, Campos & Klinnert (1985) found that most infants crossed toward a smiling parent and almost none crossed toward a fearful expression. Walden & Ogan (1988) traced referencing as a developing skill across the first two years, not a single switch.

EVIDENCE AND CAUSAL ASSESSMENT

The visual-cliff study is an experimental manipulation of the adult's expression with a physical setup that makes the drop ambiguous at a particular depth. That is stronger identification than a correlational "looks at parent more." It is still a laboratory task: the cliff, the toy, and the posed face are not a household. Walden & Ogan (1988) is an independent laboratory program (different lab, developing skill across the first two years, ambiguous toys rather than a cliff) — a conceptual replication of referencing under uncertainty, not a household-attachment study.

EFFECT SIZE AND PERSISTENCE

The classic result is categorical in a small sample (cross / not cross), not a portable standardized effect for "social competence." Persistence as a trait is not what the studies measure.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Posed expressions, caregiver identity, infant locomotor status, and the experimenter's definition of "ambiguous" all shape the score. Stranger wariness can co-occur and is not this card.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean stranger anxiety is insecure attachment. Does not mean a parent who shows fear has injured the child. A look to an adult is information use, not a purity test of the relationship.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary presence of a trusted adult in new situations is consistent with the finding. Training parents to perform referencing, or treating a wary infant as an attachment failure, does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Sorce, Emde, Campos & Klinnert (1985), Developmental Psychology
- Walden & Ogan (1988), Child Development

Infant Statistical Speech Segmentation

Eight-month-olds can use transitional probabilities in a brief artificial-speech stream to distinguish word-like units from part-words; this is laboratory segmentation, not proof that statistics alone explain natural language.

GRADE: HIGH-LAB; LOW-TRANSFER | STATUS: STABLE | DOMAIN: LANGUAGE, LEARNING | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Fluent speech does not arrive pre-segmented. In a classic laboratory demonstration, 8-month-old infants exposed to two minutes of a continuous artificial-speech stream later distinguished syllable sequences that had been “words” in that stream from sequences that straddled word boundaries. Transitional probability was the available cue.

EVIDENCE AND CAUSAL ASSESSMENT

Saffran, Aslin & Newport (1996), Science: 8-month-olds; familiarization with a continuous stream of three-syllable nonsense words; test preference for words vs part-words. Aslin, Saffran & Newport (1998), Psychological Science, held syllable frequency constant and still found discrimination based on conditional probability — a conceptual replication of the computational claim, not a household-vocabulary study. The design manipulates the statistics of the input and measures immediate segmentation. It does not assign meaning, social reference, or later vocabulary.

EFFECT SIZE AND PERSISTENCE

The result is immediate looking/listening preference after a short exposure. Transfer to the child’s language, durability without further exposure, and contribution to real vocabulary are not established by these experiments.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Artificial language; small classic samples; laboratory attention. Adjacent Chapter 6 contingency learning is a different action–outcome claim.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean the infant already knows words in the adult sense, or that a parent must provide a statistically engineered stream. Detecting a regularity is not knowing a meaning.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary talk in a shared world is consistent with the finding. Segmentation software for infants is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Saffran, Aslin & Newport (1996), Science 274:1926–1928
- Aslin, Saffran & Newport (1998), Psychological Science 9:321–324

Still-Face Disruption of Expected Social Contingency

When a familiar adult suddenly becomes unresponsive in face-to-face interaction, infants typically protest, look away, and try to re-engage; this is a laboratory signature of expected contingency, not a diagnosis of attachment or co-regulation failure.

GRADE: HIGH-LAB; LOW-ATTACHMENT-VERDICT | STATUS: STABLE | DOMAIN: EMOTION, REGULATION | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

In the still-face procedure, a familiar adult who has been interacting normally becomes suddenly unresponsive. Infants typically reduce smiling, increase negative affect and gaze aversion, and often try to re-engage the adult. Tronick, Als, Adamson, Wise & Brazelton (1978) established the laboratory signature. Mesman, van IJzendoorn & Bakermans-Kranenburg (2009) synthesized decades of replications: the protest/re-engagement pattern is robust as a procedure effect, not as a personality score.

EVIDENCE AND CAUSAL ASSESSMENT

The design is experimental: the same adult, the same infant, a controlled change in contingency. That is stronger than a correlational "synchrony score." It is still a laboratory episode. Mesman et al. (2009) document heterogeneity by age, coding, and recovery-period measurement. The existing co-regulation card covers bidirectional regulation as a construct; this card covers one experimental disruption of that process. Sleep and attachment-intervention cards are not restated.

EFFECT SIZE AND PERSISTENCE

The classic result is a within-episode change in affect and looking, not a portable standardized "regulation trait." Persistence as a forecast of later competence is not what the paradigm measures.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Cultural display rules, infant state, and how "recovery" is coded all move the score. A difficult still-face is not institutional deprivation.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean a distressed still-face proves insecure attachment or parental failure. Distress is information about broken contingency, not a purity test. Co-regulation is not constant calm.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary, repairable interaction is consistent with the finding. Training parents to "pass" a still-face, or ranking infants by protest, does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Tronick, Als, Adamson, Wise & Brazelton (1978), Journal of the American Academy of Child Psychiatry
- Mesman, van IJzendoorn & Bakermans-Kranenburg (2009), Developmental Review

Toddler Inhibitory Control Is Context-Bound

Response suppression becomes more reliable across toddlerhood and preschool under particular cues and supports; failure to inhibit is not character, and delay-of-gratification destiny claims belong on the Watts card.

H-AGE-TREND; MODERATE-TASK; LOW-MARSHMALLOW-DESTINY | STATUS: DRAFT | DOMAIN: EXECUTIVE-FUNCTION, REGULATION

WHAT WE KNOW (Narrowest Supported Conclusion)

Early forms of stopping or delaying a prepotent response are visible in infancy and become more reliable through toddlerhood and preschool. Garon, Bryson & Smith (2008) treat inhibition as a developing component, not a switch from impulse to control. Carlson (2005) catalogued age-sensitive tasks. Carlson, Davis & Leach (2005) showed that three-year-olds wait or choose better when the reward is represented symbolically ("Less Is More") than when it is sitting there as a concrete temptation—evidence that inhibition is partly a representation problem, not a character trait.

EVIDENCE AND CAUSAL ASSESSMENT

Age trends come from batteries and longitudinal task studies. Experimental cue, language, and symbolic-distancing manipulations show that performance moves with context. That is stronger than a trait interpretation. It is not an RCT of parenting. The Watts 2018 replication remains the identification for the original strong-predictor claim; this card does not reopen it.

EFFECT SIZE AND PERSISTENCE

Population age trends are large; individual scores fluctuate with fatigue, emotion, and familiarity. Early scores have limited precision as forecasts.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Language comprehension, reward value, and adult scaffolding are confounds and supports. Play-as-EF-training is a different, weaker claim (pretend-play-educational-claims.md).

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Failure to inhibit is not bad parenting, weak character, or absence of learning. Toddlers are not expected to apply a rule across emotional and unfamiliar contexts. Waiting longer is not superior character.

WHAT ACTION IS REASONABLY JUSTIFIED

Make rules concrete and reduce unnecessary temptation; these are proportionate supports. Ranking children by inhibition tasks, or training marshmallow performance, does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Garon, Bryson & Smith (2008), Psychological Bulletin
- Carlson (2005), Developmental Neuropsychology
- Carlson, Davis & Leach (2005), Psychological Science

Instrumental Helping Is Not Mature Altruism

In the second year, children sometimes help an adult complete a simple goal; this laboratory emergence is not mature altruism, a character forecast, or a parenting score.

DE: MODERATE-EMERGENCE; LOW-CHARACTER-FORECAST | STATUS: DRAFT | DOMAIN: SOCIAL, BEHAVIOR | REPLICATION: VERIF

WHAT WE KNOW (Narrowest Supported Conclusion)

Warneken & Tomasello (2006) showed that 18-month-olds often helped an unfamiliar adult complete simple failed goals (picking up an out-of-reach object, opening a cabinet) without being asked and without apparent reward. Chimpanzees helped in a narrower subset of the same problems. The object is instrumental helping in a laboratory, not a moral trait.

EVIDENCE AND CAUSAL ASSESSMENT

Small-N laboratory experiments with a small set of goal types. Replication across labs exists for the basic helping act; interpretation as "altruism" is a further claim. Adjacent: committed-compliance-not-obedience.md is a different object (willing cooperation with a caregiver's agenda). Peer coordination (toddler-peer-interaction.md) is not this helping act. Sleep and attachment-intervention theory are not restated.

EFFECT SIZE AND PERSISTENCE

Helping rates are high for some out-of-reach tasks and lower for others. Persistence as a trait is untested by this design. A child who does not help in one session has not been shown to lack a social mind.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (laboratory behavioral task, not an observational population phenotype association).

Experimenter presence, task familiarity, and cultural expectations about assisting adults are live threats. Samples are modest and Western.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Instrumental helping is not mature altruism, conscience, or a parenting score. Failure to help is not a character defect. A helping trial is not an attachment classification.

WHAT ACTION IS REASONABLY JUSTIFIED

Treat second-year helping as a developing competence, not a moral ranking. A kindness curriculum does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Warneken & Tomasello (2006), Science

Joint Attention Is Shared Reference, Not a Language Guarantee

Joint attention is the coordinated processing of one's own attention, another's attention, and a shared referent; associations with later communication are real and not a proof that drilling gaze-following produces language.

DE: MODERATE-CONSTRUCT; LOW-ISOLATED-TRAINING | STATUS: DRAFT | DOMAIN: COGNITION, LANGUAGE | REPLICATION: VERI

WHAT WE KNOW (Narrowest Supported Conclusion)

Before a full theory of other minds, infants process information about their own attention and another person's attention together. Mundy & Newell (2007) describe joint attention as that integrated processing, supported by anterior and posterior attention systems, and as a contributor to later social cognition. Responding to joint attention and initiating it develop on overlapping but different schedules.

EVIDENCE AND CAUSAL ASSESSMENT

Mundy & Newell (2007), Current Directions in Psychological Science, is a theoretical/neural-systems synthesis, not an RCT of joint-attention training in typical samples. It draws on longitudinal and individual-difference work (including Mundy's own programs) showing developmental associations among early attention, joint attention, and later social-cognitive measures. Identification is correlational and experimental in laboratory interaction, not a population trial. Cultural organization of gaze and pointing is not erased by a cortical-network sketch.

EFFECT SIZE AND PERSISTENCE

No portable effect size for "more joint attention" as an intervention in typical development. Associations with later language are informative and not destiny.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (developmental interaction construct, not an observational population phenotype association).

Infant characteristics, caregiver behavior, language environment, and measurement (responding vs initiating) all enter the score. Autism-related joint-attention findings are a clinical literature; they do not describe ordinary missed looks.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean a parent must catch every gaze. Does not mean Hart & Risley word counts. Does not mean a missed episode of shared looking is a missed mind. Conversational-turns prediction (Gilkerson) is a different card.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary shared looking, pointing, and showing are consistent with the construct. Gaze-following drills as cognitive enhancement are not justified here.

VERIFIED SOURCES & DOI IDENTIFIERS

- Mundy & Newell (2007), Current Directions in Psychological Science (DOI 10.1111/j.1467-8721.2007.00518.x)

Looking Time Is a Measurement, Not a Window into Thought

Infant looking-time methods reliably index perceptual discrimination, novelty, and attentional salience under controlled conditions; looking duration is not a transparent readout of rich conceptual knowledge, belief, or innate theories.

GRADE: HIGH-METHOD; LOW-RICH-INTERPRETATION | STATUS: STABLE | DOMAIN: COGNITION | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

The most common behavioral technique in infant perception, cognition, language, and social research is some variant of looking time (preferential looking, habituation/dishabituation, and violation-of-expectation). A longer look indexes perceptual discrimination, novelty detection, or differential attentional allocation across stimulus displays. It does not, by itself, reveal the representational architecture or conceptual content that produced the look.

EVIDENCE AND CAUSAL ASSESSMENT

Aslin (2007), Developmental Science, articulates the formal logic of looking-time inference, demonstrating that the validity of any conclusion depends strictly on the linking hypothesis connecting the dependent measure (fixation duration) to an underlying cognitive state. Without an explicit, testable linking hypothesis, looking-time differences are fundamentally underdetermined.

Haith (1998), Infant Behavior and Development, provides the foundational empirical and theoretical critique of nativist overreach ("Who put the cog in infant cognition?"), showing that lower-level perceptual properties (e.g., visual contrast, motion vectors, spatial boundaries, and ocular salience) routinely account for looking preferences previously attributed to sophisticated mental models of object permanence, arithmetic, or morality.

ManyBabies Consortium (Frank et al. 2020), Advances in Methods and Practices in Psychological Science, empirically quantified looking-time measurement dynamics across 2,329 infants and 67 laboratories. The multi-lab investigation confirmed overall robust measurement validity while identifying systematic method-dependent effect-size variation (central fixation vs. single-screen vs. head-turn preference procedure), developmental age gradients, and substantial between-infant and test-retest measurement variance.

The identification is methodological and measurement-focused: looking duration indexes perceptual and attentional engagement. It does not provide empirical license for treating looking time as a direct readout of unobserved propositional thought.

EFFECT SIZE AND PERSISTENCE

There is no portable universal effect size for "what a look means." Across laboratories, infant looking preferences yield moderate-to-large effects for basic perceptual and novelty contrasts under controlled conditions (ManyBabies 2020 observed meta-analytic Cohen's d = 0.35 to 0.45 depending on method implementation), but these reflect immediate, display-specific attentional contrasts. They do not constitute an enduring cognitive asset or predict individual long-term developmental intelligence.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (methodological measurement property, not an observational population phenotype association).

Within-session laboratory confounding is substantial: infant state (alertness, fussiness, fatigue), sensory adaptation, trial order effects, visual saliency mismatches, and regression to the mean routinely distort looking metrics. Multi-lab data demonstrate that single-lab looking-time studies with sample sizes of N = 12–16 per cell are underpowered and subject to publication bias and analysis flexibility.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean infant looking-time research is invalid or should be discarded. Does not mean infants lack perceptual capabilities. Does not license treating a 3-second laboratory look at an "impossible" event as proof that an infant possesses innate physical mechanics, arithmetic theories, or moral judgment. Detecting a visual discrepancy is not equivalent to holding a conceptual theory.

WHAT ACTION IS REASONABLY JUSTIFIED

Parents and educators should ignore commercial products and popular claims asserting that infant looking patterns prove high-level cognitive instruction or infant "genius." Experimental claims about what an infant "knows" must always be qualified by naming the specific behavioral index (looking duration) and the alternative perceptual interpretations.

VERIFIED SOURCES & DOI IDENTIFIERS

- Aslin (2007), Developmental Science (DOI: 10.1111/j.1467-7687.2007.00563.x (https://doi.org/10.1111/j.1467-7687.2007.00563.x))

- Haith (1998), Infant Behavior and Development (DOI: 10.1016/s0163-6383(98)90001-7 (https://doi.org/10.1016/s0163-6383(98)90001-7))

- ManyBabies Consortium / Frank et al. (2020), Advances in Methods and Practices in Psychological Science (DOI: 10.1177/2515245919900809 (https://doi.org/10.1177/2515245919900809))

Neighborhood Moves Have Age-Dependent Later Effects

Moving to a lower-poverty neighborhood before adolescence increased later college attendance and adult earnings in Moving to Opportunity; this is an adult-economic finding, not a toddler-development law, and moving is not universally beneficial.

TER-EARNINGS; LOW-TO-MODERATE-EARLY-DEV; LOW-EVERY-CHARACTERISTIC | STATUS: DRAFT | DOMAIN: ECOLOGY, POLICY | R

WHAT WE KNOW (Narrowest Supported Conclusion)

Housing is a bundle: stability, crowding, affordability, safety, toxins, mobility, and neighborhood opportunity. Chetty, Hendren & Katz (2016) reanalyzed Moving to Opportunity: children who moved to lower-poverty neighborhoods before age 13 had higher later college attendance and adult earnings; moves in adolescence produced smaller or slightly negative effects, plausibly reflecting disruption. Earlier MTO reports found few detectable effects on contemporaneous achievement—near-term developmental scores and long-run economic outcomes can diverge.

EVIDENCE AND CAUSAL ASSESSMENT

MTO is a randomized voucher experiment. Effects belong to the intervention package and context, not to a broad "neighborhood quality" construct. BEIP (severe-institutional-deprivation.md) is severe deprivation, not ordinary housing mobility. Childcare quality is a different exposure.

EFFECT SIZE AND PERSISTENCE

The headline MTO estimate for young movers is about 35% higher individual earnings in adulthood—not a toddler test-score effect. Persistence here is life-course economic, not an infant cognition claim.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (randomized housing voucher experiment, not an observational parenting phenotype).

Age at move, disruption, take-up, school and peer environments, and pollution all matter. Neighborhood effects in the broader observational literature are typically modest.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Moving is not universally beneficial. Place does not determine destiny. Families cannot individually solve hazardous housing and segregation. A later earnings gain is not proof that toddlers in higher-poverty neighborhoods lack parental care.

WHAT ACTION IS REASONABLY JUSTIFIED

Prioritize stable, affordable, healthy housing and hazard reduction. Evaluate mobility policies for both benefits and disruption, by child age and domain.

VERIFIED SOURCES & DOI IDENTIFIERS

- Chetty, Hendren & Katz (2016), American Economic Review

Neonatal Cardiopulmonary Transition

Birth reorganizes gas exchange and circulation after placental separation; shunt closure is functional, not instant destiny, and a difficult transition is not a personality forecast.

GRADE: HIGH-SEQUENCE; MODERATE-TIMING | STATUS: STABLE | DOMAIN: BIRTH, REGULATION | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

After placental separation, gas exchange must move from placenta to lungs. Lung aeration and oxygenation lower pulmonary vascular resistance; loss of the low-resistance placental circuit raises systemic vascular resistance. Those pressure changes drive functional closure of the foramen ovale, ductus arteriosus, and ductus venosus. Metabolic and endocrine support (cortisol, catecholamines, glucose) is part of the same transition, not a separate personality system. Similarly, transitional bilirubin metabolism and physiological neonatal jaundice (managed when clinically indicated with newborn phototherapy / photo therapy under blue light) reflect temporary hepatic enzyme clearance, not systemic illness.

EVIDENCE AND CAUSAL ASSESSMENT

Morton & Brodsky (2016) review fetal circulation (ductus venosus, foramen ovale, ductus arteriosus), lung-fluid clearance, and the postnatal rise in systemic and fall in pulmonary resistance. Hillman, Kallapur & Jobe (2012) independently describe the same sequence: lung aeration as the initiator, falling pulmonary and rising systemic resistance, and functional rather than instantaneous anatomical shunt closure. Functional flap closure of the foramen ovale follows the left-atrial pressure rise; ductal constriction is oxygen-sensitive. Human experiments cannot ethically withhold the first breath; illness, prematurity, and asphyxia are natural contrasts and are confounded by treatment.

EFFECT SIZE AND PERSISTENCE

The immediate outcome is oxygenation and a viable circulation. Transient respiratory or shunt findings do not inherit a long-run cognitive or temperament effect size. Persistence of later risk must be measured separately and is strongly modified by gestational age, illness, and subsequent care.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Preterm birth and cesarean without labor alter cortisol and catecholamine patterns; asphyxia changes the physiology of transition. Diagrams that treat “closure” as a single instant oversimplify functional versus anatomical change.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

A difficult first breath or a temporarily open shunt is not a developmental sentence. A smooth transition is not evidence of superior future cognition or character. Caregivers do not replace the placenta as an organ. Furthermore, neonatal cardiopulmonary transition and transient physiological neonatal jaundice (managed routinely when indicated with newborn phototherapy / photo therapy blue light irradiance to isomerize unconjugated bilirubin) reflect acute physiological adaptation and hepatic enzyme maturation, not cranial deformities, developmental disability, or a temperament forecast.

WHAT ACTION IS REASONABLY JUSTIFIED

Trained assessment of breathing, warmth, and resuscitation when indicated is justified. Using neonatal transition as a personality origin story is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Morton & Brodsky (2016), Clinics in Perinatology (PMC4987541)
- Hillman, Kallapur & Jobe (2012), Clinics in Perinatology

Neonatal Tongue-Tie (Ankyloglossia) and Frenotomy

Cochrane review of randomized trials finds that surgical tongue-tie release (frenotomy) modestly reduces short-term maternal nipple pain in classic anterior ankyloglossia, but shows no consistent effect on long-term breastfeeding duration or infant weight gain.

GRADE: MODERATE | STATUS: DRAFT | DOMAIN: FEEDING, HEALTH | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

A Cochrane systematic review of randomized controlled trials (oshea-2017-frenotomy) suggests that surgical division of the lingual frenulum (frenotomy) in newborns with classic anterior tongue-tie and breastfeeding difficulties produces a small short-term reduction in mothers' self-reported nipple pain, but does not demonstrate a consistent positive effect on objective infant feeding efficiency or long-term breastfeeding continuation. No randomized trial evidence supports routine laser release of "posterior" tongue-ties, upper lip-ties, or buccal (cheek) ties for general infant fussiness, reflux, or sleep.

EVIDENCE AND CAUSAL ASSESSMENT

- Cochrane Systematic Review (oshea-2017-frenotomy, 5 RCTs, N=302 infants): Pooled analysis of three studies (N=212) showed a small reduction in maternal nipple pain scores immediately post-procedure on a 0–10 scale. However, across the five trials, objective LATCH / IBFAT breastfeeding assessment scores showed inconsistent changes, and because control-group infants were offered frenotomy within days or weeks, trials could not establish whether frenotomy improves long-term exclusive breastfeeding rates or whether early latch pain resolves equally with skilled lactation positioning and infant oral growth.

EFFECT SIZE AND PERSISTENCE

Short-term maternal pain reduction on a 10-point visual analog scale averages roughly 0.7 to 1.2 points immediately after scissor frenotomy in classic anterior tongue-tie (oshea-2017-frenotomy). Long-term differences in breastfeeding duration at 3 and 6 months are unestablished in randomized designs.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: partially confounded (20–50%) in clinical practice because subjective outcome measures (maternal pain and perceived latch improvement) are vulnerable to placebo/expectation effects when parents pay out-of-pocket for unblinded laser procedures, and natural maturation of the newborn oral cavity over weeks 2–6 resolves many early latch difficulties without surgery.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean true, severe classic anterior tongue-tie (where the tongue tip is tethered to the lower gum ridge and cannot elevate or extend) does not exist. Rather, it means that the recent 5- to 10-fold commercial surge in multi-site oral laser releases ("posterior tongue-tie", "lip-tie", and "cheek-tie" packages) for routine newborn latch discomfort, gas, or reflux goes far beyond what randomized clinical evidence supports.

WHAT ACTION IS REASONABLY JUSTIFIED

When breastfeeding is painful or milk transfer is slow, start with hands-on positioning and latch adjustment from an evidence-based lactation specialist or pediatrician; reserve simple scissor frenotomy for clear, functionally restrictive anterior tongue-tie when conservative latch changes fail, and view multi-site laser oral surgeries for vague behavioral symptoms (reflux, colic, or future speech) with strong skepticism.

VERIFIED SOURCES & DOI IDENTIFIERS

- O'Shea et al. (2017), Cochrane Database of Systematic Reviews, Frenotomy for tongue-tie in newborn infants (oshea-2017-frenotomy)

Object Permanence Is Not a Single Staircase

Looking-time studies report sensitivity to occlusion events in young infants; usable search appears later and depends on action demands. Looking and reaching measure different layers, not one completed object concept.

DE: MODERATE-LOOKING-SENSITIVITY; LOW-COMPLETE-CONCEPT | STATUS: DRAFT | DOMAIN: COGNITION | REPLICATION: VERIF

WHAT WE KNOW (Narrowest Supported Conclusion)

Piaget's search tasks placed usable object permanence near the end of the first year because they required coordinated action. Baillargeon (1987) used a drawbridge/occlusion looking-time design: 4.5-month-olds, and some 3.5-month-olds, looked longer at an "impossible" rotation through the space of a hidden box than at a possible rotation. That is evidence of sensitivity to an occlusion event in a looking measure. It is not by itself a complete, action-ready object concept.

EVIDENCE AND CAUSAL ASSESSMENT

Baillargeon (1987), Developmental Psychology: experimental contrast of possible vs impossible events after habituation, with a control condition that lacked the box. Identification is looking-time, not search. Later commentary (including Aslin 2007 and subsequent VoE critiques) treats linking hypotheses as open. Competence and performance can diverge: an infant may look longer at an impossible event and still fail a manual search that requires means–end coordination.

EFFECT SIZE AND PERSISTENCE

The outcome is contemporaneous looking, not later physics competence. Persistence of a "concept" across tasks is exactly what this literature does not settle.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (laboratory cognitive developmental task, not an observational population phenotype association).

Salience, rotation extent, habituation criteria, and trial structure can change looking. Small classic samples. Replication and alternative perceptual accounts remain live. Task demands (looking vs reaching vs multiple objects) are not noise; they are part of the claim.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean young infants are miniature physicists. Does not mean Piaget should be discarded as a whole. Does not mean a missed peekaboo writes a cognitive fate. Core-knowledge slogans are not this card.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary play with disappearing and reappearing objects is consistent with the finding. Object-permanence toys as cognitive enhancement are not justified.

VERIFIED SOURCES & DOI IDENTIFIERS

- Baillargeon (1987), Developmental Psychology (DOI 10.1037/0012-1649.23.5.655)

Resilience Often Arises from Ordinary Adaptive Systems

Resilience in development commonly depends on ordinary systems—relationships, regulation, competence, and community resources—not rare invulnerability; adversity is not harmless.

HIGH-EXISTENCE; MODERATE-MECHANISM; LOW-COSTLESS | STATUS: DRAFT | DOMAIN: RESILIENCE, SYSTEMS | REPLICATION: V

WHAT WE KNOW (Narrowest Supported Conclusion)

Many children facing serious adversity later function within typical ranges. Masten (2001) argued that this is usually "ordinary magic": resilience processes ride on common adaptive systems—attachment relationships, cognitive competence, self-regulation, and community resources—rather than rare, heroic traits. Kauai and related cohorts are historical illustrations, not this card's identification.

EVIDENCE AND CAUSAL ASSESSMENT

Narrative synthesis of a research program, not a new RCT. Causal support for some protective processes comes from caregiving and placement trials already on other cards (BEIP; attachment-intervention sensitivity). This card does not re-estimate those packages. Sleep and screen theory are not restated.

EFFECT SIZE AND PERSISTENCE

There is no portable d for "resilience." Heterogeneity is the finding: some high-risk children do well, many remain affected, and recovery can be domain-specific.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (systemic adaptive framework, not an observational population phenotype association).

Who is counted as "resilient" depends on which outcome and which threshold. Survivorship and WEIRD-sample limits apply. Severe, chronic deprivation is a different identification (BEIP).

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Resilience does not mean adversity is harmless, that every child bounces back, or that families should be asked to compensate for missing housing, income, or safety. It does not identify "invulnerable" children.

WHAT ACTION IS REASONABLY JUSTIFIED

Strengthen ordinary supports (stable relationships, safety, learning opportunities) and reduce structural load. Searching for exceptional child traits as the policy is not justified.

VERIFIED SOURCES & DOI IDENTIFIERS

- Masten (2001), American Psychologist

Risky Outdoor Play Is Not a 0–3 Developmental Law

A systematic review finds associations between risky outdoor play and some health outcomes in children, mostly older than this atlas's window; it is not a toddler efficacy package or a license for unsupervised hazard.

-SCHOOL-AGE-ASSOCIATION; LOW-TO-MODERATE-0-3; LOW-TODDLER-PACKAGE | STATUS: DRAFT | DOMAIN: PLAY, HEALTH | REPL

WHAT WE KNOW (Narrowest Supported Conclusion)

Brussoni and colleagues (2015) systematically reviewed studies of risky outdoor play and health. The literature, largely school-age, reports associations with physical activity and some social and injury-pattern outcomes. It does not isolate risky play as a unique cause of later intelligence, self-control, or school success.

EVIDENCE AND CAUSAL ASSESSMENT

Systematic review, not an RCT of 0–3 outdoor play. Age bands, definitions of "risky," and outcomes vary. Adjacent: symbolic-pretend-play-emergence.md and pretend-play-educational-claims.md already refuse play-as-curriculum. Exploration (exploration-as-sampling.md) is a different identification. Screens are not re-theorized as outdoor's opposite.

EFFECT SIZE AND PERSISTENCE

No portable d for "risky play" in toddlers. Injury findings are mixed and definition-dependent. Persistence into later competence is largely untested in this atlas's window.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: untested-in-GI (no twin or sibling designs for 0–3 outdoor play; observational associations reflect family lifestyle selection).

School-age samples dominate. Families who allow more outdoor risk also differ in resources, neighborhoods, and supervision. Injury surveillance and play observation are not the same design.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Outdoor play is not worthless. It is not a registered 0–3 intelligence or character program. Removing all challenge is not shown to be required, and unsupervised hazard is not a developmental necessity.

WHAT ACTION IS REASONABLY JUSTIFIED

Do not import school-age reviews as toddler laws. Safety and opportunity are both live Art. A risky-play product does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Brussoni et al. (2015), International Journal of Environmental Research and Public Health

Parent-Implemented Language Interventions

Across 25 RCTs, parent-implemented language interventions produce modest gains in expressive vocabulary for young children with or at risk for language impairment; this is not a typical-development vocabulary mandate and is not a screen-time claim.

RADE: MODERATE-EXPRESSIVE-VOCAB; LOW-UNIVERSAL-TYPICAL | STATUS: DRAFT | DOMAIN: LANGUAGE | REPLICATION: VERIFI

WHAT WE KNOW (Narrowest Supported Conclusion)

Because early language is learned in caregiver–child interaction, some interventions train parents rather than only treating the child in clinic. Heidlage et al. (2020) meta-analyzed 25 RCTs (1,734 children, mean age 3.7 years) with or at risk for language impairment, including low-SES risk. Average effects: modest gains in expressive vocabulary (shared book reading g = 0.37; play/routines g = 0.50); small effects on broader expressive language; receptive language often nonsignificant. Dialogic reading (Whitehurst et al. 1988) is a named shared-book method inside that family, not a separate law.

EVIDENCE AND CAUSAL ASSESSMENT

RCTs of parent-implemented protocols. Efficacy under trial is not community effectiveness, scalability, or a universal parenting recipe (CONTRIBUTING.md intervention rule). Child speech is part of measured interaction; parent outcome measurement is often thin.

EFFECT SIZE AND PERSISTENCE

Hedges g in the small-to-moderate range for expressive vocabulary in the pooled trials. Persistence beyond post-test is unevenly reported. Dialogic reading's original samples were small.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (randomized assignment). Effects are smaller or uncertain for some autism-spectrum samples in the same meta-analysis. Families who enroll in RCTs are not all families.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean typical homes must run a language curriculum. Does not mean a missed shared book writes a life. Does not re-open Hart & Risley as a causal metric. Does not treat screen time as the inverse of these trials.

WHAT ACTION IS REASONABLY JUSTIFIED

Targeted parent-implemented support for children with or at risk for language impairment is proportionate to the RCT package. Universal optimization of every toddler's vocabulary is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Heidlage et al. (2020), Early Childhood Research Quarterly (DOI 10.1016/j.ecresq.2018.12.006)
- Whitehurst et al. (1988), Developmental Psychology (dialogic reading)
- Gilkerson et al. (2018), Pediatrics (adjacent prediction card; not this intervention)

Parental Leave Is a Heterogeneous Package, Not a Unitary Exposure

A Norwegian maternity-leave reform is associated with better long-run child outcomes; leave protects maternal physiological recovery, where extreme sleep debt predicts postpartum rage, irritability, and mood disturbance requiring clinical recognition and treatment.

MODERATE-REFORM; LOW-TO-MODERATE-0-3; LOW-UNITARY-JOB | STATUS: DRAFT | DOMAIN: ECOLOGY, POLICY | REPLICATION: V

WHAT WE KNOW (Narrowest Supported Conclusion)

Carneiro, Løken & Salvanes (2015) studied a Norwegian maternity-leave reform that extended paid leave. Children whose mothers were eligible had better long-run education and earnings outcomes than comparison cohorts. The object is a defined leave package in one institutional setting, not "employment" as a single exposure. Maternal physiological and emotional recovery also requires adequate protected time; acute sleep fragmentation, physical exhaustion, and lack of systemic support during the postpartum window frequently manifest as postpartum rage (PPD rage), post partum anger, post natal rage, acute irritability, and anxiety rather than unipolar sadness (Ou & Hall 2018). Clinical treatment and recognition require systematic screening for perinatal irritability and evidence-based interventions (CBT, sleep protection, and partner load-sharing).

EVIDENCE AND CAUSAL ASSESSMENT

- Natural-experiment / reform design (Carneiro et al. 2015), not an RCT of toddler cognition. Long-run education and earnings are not infant test scores. Adjacent: income-cash-transfer-early-development.md is a different lever (cash without mandated time). Childcare (early-childcare-quality.md) is what leave sometimes replaces. MTO is housing mobility, not leave.

- Ou & Hall (2018) conducted an integrative review of anger in postnatal mood disturbance, demonstrating that anger and postpartum rage are salient experiences across 30–40% of postpartum women, strongly triggered by extreme sleep deprivation, feeling unsupported, and perceived violations of parental expectations. Evidence-based treatment for postpartum rage and irritability emphasizes multi-modal care: clinical evaluation for underlying postpartum depression or anxiety, psychoeducation normalizing anger as a biological stress response to overload, cognitive-behavioral therapy (CBT) for maternal distress, and structural interventions ensuring uninterrupted nocturnal sleep cycles and protected parental leave.

EFFECT SIZE AND PERSISTENCE

Long-run education and earnings effects are the identification. Near-term infant scores are not this paper's object. Persistence here is the adult outcome, not a toddler battery. Postpartum rage and mood distress show marked reduction following sleep restoration, maternal leave stability, and targeted psychological care.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: robust (<20%) for the Norwegian reform policy effect (policy natural experiment / reform evaluation, not an observational family phenotype).

Eligibility, wage replacement, duration, and which parent stays home vary across countries. A night shift and a salaried leave are not one exposure. Selection into take-up remains a threat where leave is not universal. Perinatal mood and anger symptoms are confounded by baseline maternal psychiatric history and social isolation.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Employment is not a unitary harm. Leave is not a guarantee of infant cognition. A missing leave week is not a character defect, and a long leave is not a proven IQ program. Does not mean postpartum rage or maternal exhaustion reflects a personal moral failure, lack of maternal affection, or a character defect; these are biological indicators of physical depletion, sleep disruption, and unbuffered systemic overload. Does not mean postpartum anger requires punitive judgment rather than compassionate clinical screening and treatment support.

WHAT ACTION IS REASONABLY JUSTIFIED

Treat leave as an institutional package. Families cannot individually legislate eligibility. Incorporating explicit screening for postpartum rage and irritability into routine 6-week obstetric and well-child pediatric visits is clinically justified. Providing timely evidence-based treatment (CBT, physician consultation, and structural nocturnal sleep protection) addresses maternal suffering directly.

VERIFIED SOURCES & DOI IDENTIFIERS

- Carneiro, Løken & Salvanes (2015), Journal of Political Economy
- Ou & Hall (2018), Birth, DOI: 10.1111/birt.12356 (https://doi.org/10.1111/birt.12356)

Paternal Age and De Novo Mutation Rate

De novo single-nucleotide mutations in offspring increase linearly with paternal age at conception; this establishes a biological germline mutation rate, not an individual psychiatric or cognitive forecast for a child.

HIGH-MUTATION-RATE; LOW-FORECAST | STATUS: DRAFT | DOMAIN: PRECONCEPTION, PREGNANCY, GENETICS | REPLICATION: V

WHAT WE KNOW (Narrowest Supported Conclusion)

Male spermatogenesis involves continuous germline cell divisions throughout post-pubertal life (approximately 23 divisions per year), in contrast to female oogenesis where mitotic divisions cease prior to birth. In whole-genome sequencing of parent-offspring trios, the number of de novo single-nucleotide mutations transmitted to offspring increases linearly with the father's age at conception, adding roughly 1.5 to 2.0 new mutations per year of paternal age. The overwhelming majority of these mutations occur in non-coding or functionally silent genomic regions and produce no clinical phenotype.

EVIDENCE AND CAUSAL ASSESSMENT

Kong et al. (2012), Nature, sequenced 78 Icelandic parent-offspring trios and demonstrated that paternal age at conception accounted for the dominant share of variation in offspring de novo mutation counts, estimating an increase of approximately 2.0 mutations per year of paternal age, while maternal age contributed no significant linear increase.

Jónsson et al. (2017), Nature, independently replicated and substantially scaled these findings across 1,548 Icelandic trios (identifying 108,778 de novo mutations). The multi-generational analysis established a precise genome-wide paternal increase of 1.51 mutations per year of father's age (95% CI [1.45, 1.57]), alongside a smaller maternal effect of 0.37 mutations per year (95% CI [0.32, 0.43]).

The causal mechanism is biological and well-established: continuous DNA replication errors and progressive replication-independent DNA damage in spermatogonia drive the mutational clock.

EFFECT SIZE AND PERSISTENCE

The quantitative effect is approximately 1.51 de novo single-nucleotide mutations per year of paternal age. Across a 15-year difference in father's age (e.g., age 20 vs. age 35), an offspring inherits roughly 23 additional de novo point mutations out of 3.2 billion base pairs. Because over 98% of the human genome is non-coding and only a tiny fraction of coding mutations are deleterious, absolute clinical risk for specific neurodevelopmental conditions remains very low.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a for the germline biological mutation rate (measured directly via whole-genome sequence trios).

In contrast, observational epidemiological associations between advanced paternal age and child psychiatric outcomes (autism, schizophrenia) are largely confounded (>50%) by delayed reproductive selection, parental subclinical psychiatric traits, assortative mating, and socioeconomic indicators. Sibling-comparison studies demonstrate that population-level psychiatric risk shrinks substantially when comparing siblings born to the same father at different ages.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean that men in their late 30s or 40s should avoid having children out of fear of severe genetic disease. Does not mean older fathers produce children with impaired intelligence. A biological fact about germline replication does not license clinical alarmism or eugenic reproductive ranking.

WHAT ACTION IS REASONABLY JUSTIFIED

Clinical genetic counselors should provide accurate absolute risk perspectives rather than inflated relative risk statistics when prospective parents inquire about paternal age. Routine whole-exome or whole-genome screening of asymptomatic children based solely on paternal age is not clinically indicated.

VERIFIED SOURCES & DOI IDENTIFIERS

- Kong et al. (2012), Nature (DOI: 10.1038/nature11396 (https://doi.org/10.1038/nature11396))
- Jónsson et al. (2017), Nature (DOI: 10.1038/nature24018 (https://doi.org/10.1038/nature24018))

Physical Punishment Is Not a Developmental Necessity

Prospective syntheses associate spanking with worse child behavior and find no reliable positive developmental benefit; this is not a claim that every instance produces a damaged child.

MODERATE-TO-HIGH-NOT-NECESSARY; MODERATE-EFFECT-SIZE | STATUS: DRAFT | DOMAIN: BEHAVIOR, FAMILY | REPLICATION: V

WHAT WE KNOW (Narrowest Supported Conclusion)

Physical punishment (spanking and related acts) is consistently associated with poorer behavioral outcomes in prospective research. Gershoff & Grogan-Kaylor (2016) meta-analyzed spanking and child outcomes and found no evidence of a reliable positive developmental benefit, with associations in the direction of more, not fewer, behavior problems. The identification is stronger as "not necessary / not beneficial" than as a precise universal causal effect size.

EVIDENCE AND CAUSAL ASSESSMENT

Prospective longitudinal studies, cross-lagged models, and meta-analysis. Child behavior can evoke harsh responses (child-to-parent). A later contrary meta-analysis arguing near-zero average effects of "customary spanking" is a live identification dispute; it does not establish developmental benefit. This card does not reopen sleep-training or attachment-intervention theory.

EFFECT SIZE AND PERSISTENCE

Many associations are small in cross-lagged analyses. Consistency of direction and absence of demonstrated benefit matter more than a portable d. Effects may accumulate with frequency and severity.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: partially confounded (20–50%) (child aggressive temperament evokes parental harshness; twin-differences models show persistent negative associations after genetic control).

Cultural normativeness may moderate, but does not convert spanking into a developmental requirement.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean every parent who has used physical punishment will produce a damaged child. Does not mean all nonphysical consequences are harmless. Does not mean toddlers' failed inhibition (Chapter 11) is a moral failure requiring force.

WHAT ACTION IS REASONABLY JUSTIFIED

Physical punishment should not be recommended as an evidence-based developmental tool. Institutional responses should pair limits with practical support, not parent-blaming.

VERIFIED SOURCES & DOI IDENTIFIERS

- Gershoff & Grogan-Kaylor (2016), Journal of Family Psychology

The Placenta Mediates Fetal Dose

Maternal exposure is not equivalent to fetal dose; the placenta transports, metabolizes, restricts, or facilitates substances, including via an incomplete glucocorticoid buffer.

ADE: MODERATE-TO-HIGH-MECHANISM; LOW-CHILD-FORECAST | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFI

WHAT WE KNOW (Narrowest Supported Conclusion)

The placenta is a transient organ of exchange, metabolism, and endocrine signalling. Maternal blood contents are not a fetal dose. Placental 11β-HSD2 inactivates much maternal cortisol; the buffer is real and incomplete.

EVIDENCE AND CAUSAL ASSESSMENT

Burton & Fowden (2015) review transport, uterine-artery remodelling, endocrine modulation of maternal metabolism, and the placenta as a selective barrier. Seckl & Holmes (2007) review 11β-HSD2 as a fetoplacental glucocorticoid barrier: animal knockout/inhibition and human HSD11B2 mutations associate with reduced birthweight; this is mechanism plus selected human genetics, not a population stress-to-IQ path (see the stress card).

EFFECT SIZE AND PERSISTENCE

A placental enzyme level or weight is contemporaneous physiology. Lasting impairment does not follow from the existence of a pathway.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (biological placental transport physiology, not an observational population phenotype association).

Placenta sampled at birth; reverse causation; maternal–fetal genotype; over-translation of programming animal models.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean the mother “sends” every feeling or chemical to the child, or that placental markers are a personality test.

WHAT ACTION IS REASONABLY JUSTIFIED

Clinical attention to placental insufficiency and known teratogens is justified. Individual purification or guilt about ordinary maternal states is not licensed by this mechanism.

VERIFIED SOURCES & DOI IDENTIFIERS

- Burton & Fowden (2015), Phil. Trans. R. Soc. B
- Seckl & Holmes (2007), Nat. Clin. Pract. Endocrinol. Metab.

Positional Plagiocephaly and Cranial Orthotic Helmet Therapy

Randomized controlled trial evidence demonstrates that custom cranial orthotic helmets provide no added skull-rounding benefit over natural course for moderate-to-severe infant plagiocephaly by age 2.

TURAL-COURSE; LOW-FOR-SEVERE-SYNDROMIC-CRANIOSYNOSTOSIS | STATUS: DRAFT | DOMAIN: PHYSICAL-DEVELOPMENT, HEAL

WHAT WE KNOW (Narrowest Supported Conclusion)

Rigorous pragmatic randomized controlled trial evidence shows that custom cranial orthotic helmets (costing $2,000 to $3,500) provide no clinically meaningful or statistically significant advantage in skull shape recovery over the natural course (wait-and-see with physical repositioning and tummy time) by age 2 in infants with moderate-to-severe nonsynostotic positional plagiocephaly or brachycephaly. Spontaneous skull re-rounding occurs at statistically indistinguishable rates in both groups once infants achieve independent sitting and upright motor control. Prescribing a helmet for flat spot or for correcting plagiocephaly / brachycephaly routinely addresses parental cosmetic worry rather than underlying neurological risk.

EVIDENCE AND CAUSAL ASSESSMENT

- *van Wijk et al. (2014), BMJ** (The HEADS Study*, n=84): Pragmatic randomized controlled trial in the Netherlands enrolling healthy infants aged 5 to 6 months with moderate to severe positional skull deformation (plagiocephaly or brachycephaly). Infants were randomized to either 6 months of active helmet therapy (custom-molded passive orthosis worn 23 hours per day until age 12 months) or natural course without orthosis. The primary outcome was objective skull shape at 2 years of age measured by blinded plagiocephalometry (oblique diameter difference index [ODDI] for plagiocephaly and cranioproportional index [CPI] for brachycephaly).

- Primary RCT Findings: At 2 years of age, there was no significant difference in skull shape improvement between the helmet therapy group and the natural course group. The mean difference in ODDI was 0.2 (95% CI, -1.6 to 1.9, p=0.86), and the mean difference in CPI was 0.2 (95% CI, -1.7 to 2.1, p=0.80). Equal proportions of infants achieved full recovery: 25.6% in the helmet therapy group versus 22.5% in the natural course group (relative risk 1.14; 95% CI, 0.55 to 2.36).

- Adverse Effects & Burdens: In the helmet therapy arm, 100% of parents reported one or more adverse side effects during the 6-month wearing period: skin irritation/pressure sores (96%), augmented sweating (71%), unpleasant odor (76%), displacement of the helmet (60%), and skin pain/discomfort (33%).

EFFECT SIZE AND PERSISTENCE

The measured effect size of helmet therapy compared to natural course on objective cranial asymmetry at 2 years is essentially zero (standardized mean difference d ≈ 0.02). Spontaneous cranial remodeling is substantial across both arms between 6 and 24 months as the brain expands against gravity, skull bones thicken, and infants spend majority daytime hours sitting, crawling, and upright. Long-term follow-ups show no differences in cognitive, developmental, or motor outcomes between children treated with helmets versus those managed with observation.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (pragmatic randomized controlled trial testing a medical device intervention, not an observational heritability phenotype).

- Observational Confounding in Prior Commercial Literature: Non-randomized cohort studies showing "dramatic improvements" in helmet clinics compared treated infants at 6 months to their baseline at 4 months, confounding the helmet with the natural, spontaneous re-rounding that occurs in almost all non-syndromic infants as motor milestones emerge.

- Exclusion Boundaries: The trial explicitly excluded infants with true synostosis (premature cranial suture fusion requiring surgical repair), severe torticollis unresponsive to physical therapy, or dysmorphic syndromes.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

This evidence does not mean that positional flattening does not exist or that parents should avoid the supine "Back to Sleep" position for infant sleep. Back sleeping reduces Sudden Infant Death Syndrome (SIDS) by over 50% and is non-negotiable. Positional flattening is a benign cosmetic condition caused by prolonged external pressure on soft neonatal cranial bones; it does not compress the brain or impair neurocognitive development. Furthermore, this does not mean that premature fusion of cranial sutures (craniosynostosis) can be ignored; craniosynostosis is a structural pathological condition requiring pediatric surgical evaluation.

WHAT ACTION IS REASONABLY JUSTIFIED

Clinicians and parents can avoid expensive, uncomfortable cranial orthotic helmets ($2,000 to $3,500) for standard positional plagiocephaly and brachycephaly. The evidence strongly supports conservative, non-invasive management:

1. Maintain firm supine sleep for SIDS prevention.

2. Provide frequent, supervised daytime tummy time while awake from week one to relieve occipital pressure and strengthen neck and trunk musculature.

3. Alternate head position during naps and crib resting, and vary arm positions during feedings.

4. If congenital or positional muscular torticollis (restricted neck rotation) is present, initiate pediatric physical therapy early rather than relying on a passive helmet.

VERIFIED SOURCES & DOI IDENTIFIERS

- van Wijk, R. M., van Vlimmeren, L. A., Groothuis-Oudshoorn, C. G. M., Van der Ploeg, C. P. B., IJzerman, M. J., & Boere-Boonekamp, M. M. (2014). Helmet therapy in infants with positional skull deformation: randomised controlled trial. BMJ, 348, g2741. DOI: 10.1136/bmj.g2741 (https://doi.org/10.1136/bmj.g2741)

Preconception Nutrition Beyond Folate Is Not a Second NTD Trial

Reviews of preconception nutritional risks find a mixed, nutrient-specific evidence base; folate's High/stable NTD identification does not generalize to other preconception nutrients or to later IQ.

W-TO-MODERATE-BEYOND-FOLATE; LOW-IQ-CHECKLIST | STATUS: DRAFT | DOMAIN: PRECONCEPTION, NUTRITION | REPLICATION

WHAT WE KNOW (Narrowest Supported Conclusion)

Dean and colleagues (2014) reviewed nutritional risks and interventions in the preconception period. Evidence is nutrient-specific and often thin. Periconceptional folic acid for neural-tube defects is the strong identification — already prenatal-folate-ntd.md, status: stable for that narrow claim. Other nutrients (iron, calcium, iodine in replete populations, multivitamins as a package) do not inherit that grade.

EVIDENCE AND CAUSAL ASSESSMENT

Narrative/systematic review of mixed trials and observational studies, not a new RCT. Adjacent: prenatal-folate-ntd.md remains the NTD card; do not re-estimate it. prenatal-iodine-thyroid.md is pregnancy deficiency, not a preconception IQ program. Paternal-age mutation rate (paternal-age-de-novo-mutations.md) is a different preconception object. Sperm-epigenetic lifestyle stories remain excluded as a standalone card. Sleep and attachment are not restated.

EFFECT SIZE AND PERSISTENCE

Folate's NTD risk reduction is large and already graded High/stable. Other reviewed interventions have smaller, inconsistent, or baseline-dependent effects. Persistence to childhood cognition is generally untested or confounded.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (nutritional epidemiology and trials, not an observational population phenotype).

Baseline deficiency determines marginal benefit. Trials in deficient populations do not license supplements in replete ones. Observational diet–IQ links mix education, income, and health care.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Preconception biology is not parental control. A missed supplement is not a script for later intelligence. Folate's success is not a license for every nutrient. Animal epigenetic marking is not a childhood.

WHAT ACTION IS REASONABLY JUSTIFIED

Keep folate's NTD identification. Do not sell a general preconception brain program. Nutrient claims beyond folate need their own designs.

VERIFIED SOURCES & DOI IDENTIFIERS

- Dean et al. (2014), Reproductive Health

Prenatal Alcohol and Fetal Alcohol Spectrum Disorders

Prenatal alcohol is causally implicated in FASD spanning growth, facial, and neurobehavioral outcomes; detected exposure is not equivalent to FASD, and not all neurobehavioral problems are alcohol-caused.

GRADE: HIGH-FASD; LOW-ANY-SIP | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Prenatal alcohol exposure is causally implicated in a clinical continuum (FASD) of growth restriction, characteristic facial features, and neurobehavioral impairment. Severity is dose- and timing-dependent. A detected exposure is not a diagnosis; a neurobehavioral problem is not proof of alcohol causation.

EVIDENCE AND CAUSAL ASSESSMENT

Hoyme et al. (2016) update diagnostic criteria from evaluation of more than 10,000 children and NIAAA-linked epidemiology (CIFASD, CoFASP): documented prenatal alcohol exposure, facial features, growth, CNS, and neurobehavior. This is clinical teratology and consensus diagnosis, not a randomized exposure trial. Human causal inference rests on dose, pattern, dysmorphology, and exclusion of competing causes.

EFFECT SIZE AND PERSISTENCE

FASD persistence is often substantial. Unmeasured low-level exposure does not inherit that effect size.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (teratological toxicology, not an observational population phenotype association).

Exposure under-report; polysubstance use; SES and nutrition; stigma; diagnostic substitution.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean every drink produces FASD, or that every developmental difficulty is alcohol-caused, or that affected children have a fixed fate.

WHAT ACTION IS REASONABLY JUSTIFIED

Avoiding alcohol in pregnancy and supporting affected children are justified. Shame as a developmental theory is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Hoyme et al. (2016), Pediatrics

Prenatal Auditory Discrimination and Newborn Preference

Late-pregnancy speech exposure can shape newborn auditory preference and neonatal ERP memory traces; this is sensory discrimination and neural encoding, not evidence of complex prenatal cognition or lasting cognitive advantage.

ERATE-PREFERENCE; LOW-LASTING-ADVANTAGE | STATUS: DRAFT | DOMAIN: PREGNANCY, LANGUAGE, PERCEPTION | REPLICATIO

WHAT WE KNOW (Narrowest Supported Conclusion)

Third-trimester fetuses hear maternal speech and external acoustic stimuli through the abdominal wall and amniotic fluid. Newborns show behavioral operant preference for speech passages recited repeatedly by their mother in the final weeks of gestation over novel passages. Parallel electrophysiological recordings demonstrate that prenatal exposure to specific auditory patterns leaves detectable neural memory traces (mismatch negativity) in the neonatal brain. This reflects sensory-neural attunement and short-horizon auditory familiarity, not complex symbolic learning or lasting intellectual advancement.

EVIDENCE AND CAUSAL ASSESSMENT

DeCasper & Spence (1986), Infant Behavior and Development, demonstrated that infants whose mothers read a specific passage (e.g., The Cat in the Hat) twice daily during the last six weeks of pregnancy modified their non-nutritive sucking burst-pause patterns to hear the familiar story over a novel control story. This experimental manipulation showed that acoustic properties (prosodic rhythm, pitch contours) are registered prenatally and act as reinforcing stimuli postnatally.

Partanen et al. (2013), Proceedings of the National Academy of Sciences, provided converging neural evidence by exposing fetuses to a repeated pseudoword sequence from gestational week 29 until birth. At birth, exposed newborns displayed significant event-related potential responses (mismatch negativity) to structural pitch modifications in the sequence, which correlated with the volume of prenatal exposure and persisted at 4 months, confirming a neural trace of prenatal auditory experience.

The identification is sensory-attunement and neural encoding: late fetal auditory systems encode salient acoustic regularities. It does not provide empirical evidence for conceptual instruction in utero.

EFFECT SIZE AND PERSISTENCE

The behavioral preference is documented immediately post-birth (first 2–4 days of life) in operant sucking designs. Electrophysiological traces remain detectable into early infancy (up to 4 months in Partanen 2013), but there is zero empirical evidence that prenatal auditory exposure produces measurable gains in vocabulary size, grammatical competence, or school-age IQ.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (laboratory sensory encoding and operant preference, not an observational population phenotype association).

Small classic sample sizes and procedural sensitivity characterize the behavioral sucking literature. Postnatal caregiver speech immediately continues the acoustic familiarity, confounding long-term naturalistic follow-up. Maternal voice preference is also reinforced by postnatal feeding and skin contact.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean reading or playing music to a fetus accelerates child intelligence or builds a "prenatal genius." Does not mean fetuses comprehend language or understand story narratives. A newborn turning toward a mother's familiar voice is an index of evolutionary sensory attunement, not an educational head start.

WHAT ACTION IS REASONABLY JUSTIFIED

Talking, singing, and reading during pregnancy are natural, emotionally grounding human behaviors that require no specialized commercial devices. Parents should ignore "fetal learning" speakers, specialized sound belts, and prenatal curricula marketed to enhance intelligence.

VERIFIED SOURCES & DOI IDENTIFIERS

- DeCasper & Spence (1986), Infant Behavior and Development (DOI: 10.1016/0163-6383(86)90025-1 (https://doi.org/10.1016/0163-6383(86)90025-1))

- Partanen et al. (2013), PNAS (DOI: 10.1073/pnas.1302159110 (https://doi.org/10.1073/pnas.1302159110))

Prenatal Developmental Timing Is Process-Specific

The same prenatal exposure can have different effects depending on dose, route, and the developmental process underway; pregnancy is not a single window.

GRADE: MODERATE-PRINCIPLE; HIGH-NO-SINGLE-WINDOW | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Susceptibility to a prenatal exposure varies with developmental stage, dose, and the process underway. Organogenesis is more vulnerable to structural malformation than later fetal growth; the same agent is not one “prenatal effect.”

EVIDENCE AND CAUSAL ASSESSMENT

Wilson’s principles, still treated as foundational in later reviews (Ujházy et al. 2012), state that teratogenic susceptibility varies with developmental stage, that access to embryonic tissues matters, and that manifestations increase with dose from no-effect to lethal. Human application rests on clinical timing (e.g. organogenesis vs later growth/functional injury), not on randomized timed exposures.

EFFECT SIZE AND PERSISTENCE

There is no universal effect size. The unit is exposure × window × process × outcome. Persistence must be measured for each pairing; a sensitive period is not an inevitable-injury window.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (embryological organogenesis windows, not an observational population phenotype association).

Model-system translation; sparse direct human observation of early embryos; dating error; selection into ART and clinical series.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean one missed week “ruins” development, or that every prenatal difference is a malformation.

WHAT ACTION IS REASONABLY JUSTIFIED

Public-health timing for known teratogens and periconceptional folate is justified. Treating pregnancy as a single optimization window is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Ujházy et al. (2012), Interdisciplinary Toxicology (Wilson’s principles)

Prenatal Environmental Mixtures and Placental Markers

Mixture studies can associate phthalates and related chemicals with placental biomarkers; this does not establish that typical consumer-level detection causes permanent child injury.

GRADE: LOW-ASSOCIATION; VERY-LOW-PERMANENT-INJURY | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Many chemicals can reach or alter the placenta. In SEPAGES, selected phthalate metabolites (notably MBzP) were associated with lower placental weight and related markers. Phenol mixtures in the same analysis were not associated. This is a placental-marker claim, not proven child neurodevelopmental injury from typical consumer detection.

EVIDENCE AND CAUSAL ASSESSMENT

Philippat et al. (2024), SEPAGES (France): repeated pooled urines (median 21 samples/trimester), linear models and BKMR. MBzP associated with lower placental weight (about −20 g and −17 g per ln-unit in second and third trimesters) and placental-to-fetal ratio; MnBP with lower vascular resistance; DiNP associations mainly in male infants. No overall phenol-mixture association. Observational environmental epidemiology; not randomized, not genetically informed.

EFFECT SIZE AND PERSISTENCE

Placental weight shifts are small relative to clinical pathology. Child neurodevelopment was not the outcome here. Persistence and reversibility are unestablished.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (environmental toxicological biomarker study, not an observational population phenotype association).

SES and housing; multiple-exposure; sex interaction; mechanism-to-outcome leap; single-cohort replication still thin.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean every detected chemical is an injury, or that parents must purify a household to protect a child's future.

WHAT ACTION IS REASONABLY JUSTIFIED

Institutional chemical-policy and product standards can be discussed at the system level. Individual purity projects are not licensed by these associations.

VERIFIED SOURCES & DOI IDENTIFIERS

- Philippat et al. (2024), Human Reproduction (SEPAGES)

Periconceptional Folic Acid and Neural-Tube Defects

Randomized and systematic-review evidence shows periconceptional folic acid substantially reduces neural-tube-defect risk; this does not generalize to all anomalies or to intelligence.

DE: HIGH-NTD; LOW-OTHER-ANOMALIES-COGNITION | STATUS: STABLE | DOMAIN: PREGNANCY, NUTRITION | REPLICATION: VERI

WHAT WE KNOW (Narrowest Supported Conclusion)

Daily periconceptional folic acid (alone or with other vitamins) reduces neural-tube-defect-affected pregnancy relative to no folate. The Cochrane synthesis of five trials reports RR 0.31 (95% CI 0.17–0.58; 6,708 births; high-quality evidence). This is a structural embryonic outcome, not a general intelligence or "optimal development" claim.

EVIDENCE AND CAUSAL ASSESSMENT

- Landmark RCT: MRC Vitamin Study (1991) established prevention of NTD recurrence with periconceptional folic acid.

- Independent first-occurrence RCT: Czeizel & Dudás (1992), NEJM, Hungarian periconceptional multivitamin trial; no NTD in the folic-acid arm versus six in the control arm among women without a prior NTD pregnancy.

- De-Regil et al. (2015) Cochrane review: five RCTs, 6,708 births, RR 0.31 (0.17–0.58) for NTDs; recurrence RR 0.34 (0.18–0.64). Dose subgroup (0.4 mg vs higher) did not change the conclusion. Cochrane new_version on this record is a successor review, not a retraction.

- WHO antenatal-care guidance recommends daily iron and folic acid as a public-health measure; that guideline is adjacent framework, not this card's identification.

EFFECT SIZE AND PERSISTENCE

Large relative-risk reduction for NTD. Persistence is binary at the structural level (defect present or not). There is no license to carry this effect size into IQ, school achievement, or all birth defects.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a for NTD. Baseline fortification and population folate status change the absolute benefit. Trials are older and often recurrence-focused; first-occurrence evidence is now represented by Czeizel & Dudás as well as the Cochrane pool. Adherence, timing (must cover neural-tube closure), and access are implementation issues, not reasons to doubt the core RCT result.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean perfect supplementation guarantees typical development, that an NTD implies parental nutritional failure in every case, or that folic acid is a general brain-optimization tool.

WHAT ACTION IS REASONABLY JUSTIFIED

Population fortification and periconceptional folic acid as recommended antenatal care are justified on NTD-prevention grounds. Supplement maximalism and cognitive-enhancement framing are not.

VERIFIED SOURCES & DOI IDENTIFIERS

- De-Regil et al. (2015), Cochrane Database of Systematic Reviews
- MRC Vitamin Study Research Group (1991), Lancet
- Czeizel & Dudás (1992), New England Journal of Medicine

Prenatal Iodine, Maternal T4, and Neurodevelopment

Severe early iodine deficiency can disrupt neurodevelopment via maternal T4; late correction does not rewind early corticogenesis. This is not a brief for supplement maximalism in replete populations.

GRADE: HIGH-SEVERE; LOW-MILD-REPLETE | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Before mid-gestation the fetal brain depends on maternal T4. Severe iodine deficiency causes endemic cretinism (neurological injury that later iodine does not fully rewind). Milder early hypothyroxinemia is associated with altered cortical organization in experimental and some human data; that is not a license for maximal supplementation in iodine-replete settings.

EVIDENCE AND CAUSAL ASSESSMENT

Zimmermann et al. (2008) review iodine-deficiency disorders: preventable mental impairment at the severe end; salt iodization as a population measure. Morreale de Escobar et al. (2004) show maternal T4 reaches the fetal compartment early; locally generated T3 in cortex reaches adult values by mid-gestation; experimental early hypothyroxinemia disrupts neuronal migration.

EFFECT SIZE AND PERSISTENCE

Severe deficiency: large, lasting neurological injury. Mild deficiency: smaller and contested. Persistence of severe injury is structural; it is not “undone” by later correction of the same process.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (severe deficiency epidemiology and endocrine physiology, not an observational population phenotype association).

Population iodine status; over-translation of endemic data to replete settings; confounding of mild hypothyroxinemia with other pregnancy risks.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not justify supplement maximalism, or treating every learning difficulty as maternal iodine failure.

WHAT ACTION IS REASONABLY JUSTIFIED

Population iodization and preventing severe deficiency are justified. Extra iodine as cognitive enhancement in replete populations is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Zimmermann, Jooste & Pandav (2008), Lancet
- Morreale de Escobar, Obregón & Escobar del Rey (2004), Best Pract. Res. Clin. Endocrinol. Metab.

Prenatal Maternal Stress and Anxiety

Observational prenatal stress/anxiety associations with child outcomes largely attenuate under genetically informed designs.

ICAL-RANGE; MODERATE-LARGELY-CONFOUNDED; LOW-RESIDUAL-PRENATAL | STATUS: DRAFT | DOMAIN: PREGNANCY, DEVELOPM

WHAT WE KNOW (Narrowest Supported Conclusion)

Prenatal maternal stress/anxiety is observationally associated with lower birthweight, earlier gestational age, and more offspring emotional/behavioral difficulties, but these associations attenuate substantially -- in several cases to near-zero -- once genetically informed designs (sibling comparisons, children-of-twins, intergenerational Mendelian randomization) are applied.

EVIDENCE AND CAUSAL ASSESSMENT

- MoBa cohort triangulation (sibling control, gene-environment interaction, intergenerational MR, negative control analysis; N=112,784 full sample, N=36,511 sibling subsample) found sibling-control analyses substantially attenuated associations between prenatal stress and birthweight, gestational age, and emotional/behavioral difficulties; MR showed no clear causal effect of maternal cortisol.

- UK Millennium Cohort sibling-comparison study (N=21,980 siblings) found prenatal maternal anxiety associated with child difficulties in full-cohort analysis (OR 2.1-2.7) but the association disappeared in within-family sibling comparisons (OR ~1.3, CIs crossing 1).

- Meta-analysis of genetically informed designs (adoption, sibling, children-of-twins, IVF; N>11,700 families) found no association between prenatal anxiety exposure and offspring internalizing outcomes after accounting for relatedness (r=.04), versus a small but real postnatal/concurrent effect (r=.13).

- Lancet Psychiatry children-of-twins/sibling analysis found maternal prenatal depressive symptoms' association with offspring psychopathology was ~40% explained by shared genetic factors.

EFFECT SIZE AND PERSISTENCE

Observational associations are small-to-moderate; genetically informed estimates are frequently statistically indistinguishable from zero. Concurrent (postnatal) maternal anxiety/depression retains a small but real nongenetic effect (r~.13) that observational prenatal-only estimates do not.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: largely confounded (>50%) (typical-range maternal prenatal stress associations largely attenuate under sibling and children-of-twins designs).

Largely confounded (>50%) for prenatal-specific associations across multiple independent genetically informed designs and cohorts (Norway, UK, other Nordic/twin registries). Genetic transmission and shared family environment are the dominant alternative explanations. Concurrent/postnatal parental anxiety shows less attenuation and may reflect a partially independent pathway.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Findings do not license blaming pregnant people for population-range stress or anxiety, nor claiming a documented causal fetal-programming pathway for typical stress levels.

WHAT ACTION IS REASONABLY JUSTIFIED

Support for maternal mental health during and after pregnancy is justified on its own terms (concurrent effects, maternal wellbeing), not on the basis of a demonstrated independent prenatal-stress-causes-child-harm pathway at typical exposure levels.

VERIFIED SOURCES & DOI IDENTIFIERS

- BMC Medicine (2025), MoBa triangulation study
- International Journal of Epidemiology (2017), UK Millennium Cohort sibling design
- J Am Acad Child Adolesc Psychiatry meta-analysis of genetically informed designs (2021)
- The Lancet Psychiatry (2018), children-of-twins design

Prenatal Neural Construction Is Unfinished at Birth

Human cortical development proceeds through overlapping proliferation, migration, and transient zones; the brain remains immature at birth and continues under postnatal experience.

GRADE: HIGH-STRUCTURE; LOW-COGNITION-FORECAST | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Human cortical development is a sequence of overlapping processes (proliferation, migration, formation of transient embryonic zones including the subplate). Construction is unfinished at birth; postnatal experience continues the work. Structure is not intelligence or personality.

EVIDENCE AND CAUSAL ASSESSMENT

Bystron, Blakemore & Rakic (2008) revise the 1970 Boulder Committee model with human evidence: ventricular/subventricular/intermediate/cortical-plate/marginal zones, and a distinct, functionally important subplate not recognized in 1970. This is developmental neuroanatomy, not an RCT of "brain building." Severe disruptions (e.g. congenital infection, extreme growth restriction) are contrast cases, not the typical claim.

EFFECT SIZE AND PERSISTENCE

No universal cognitive effect size. Persistence of a structural difference must be measured; many prenatal differences change form after birth.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (neuroembryology and cortical development, not an observational population phenotype association).

Indirect fetal imaging; postmortem and clinical-sample selection; species differences even in a human-focused review; postnatal confounding of later outcomes.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean the newborn brain is a blank slate, or that prenatal construction writes adult character.

WHAT ACTION IS REASONABLY JUSTIFIED

Protecting pregnancy against known severe disruptors is justified. "Prenatal brain training" is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Bystron, Blakemore & Rakic (2008), Nature Reviews Neuroscience

Maternal Tobacco Smoke and Fetal Growth

Active maternal tobacco smoking is associated with lower birthweight and higher preterm risk, with dose-response evidence; this is a birth-size/preterm claim, not a personality or IQ card.

GRADE: MODERATE | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Active maternal cigarette smoking is associated with reduced fetal growth and higher preterm risk, with a reported dose-response. This is a birth-size and gestational-duration claim. It is not a personality or IQ card.

EVIDENCE AND CAUSAL ASSESSMENT

Kramer (1987) assessed 43 determinants across 895 studies: in developed-country settings, cigarette smoking was the most important identified modifiable determinant of low birthweight, with effects on both growth restriction and gestational duration. Later biomarker work (not catalogued here as a second ID) often shows tighter dose-response with cotinine than with self-report. This remains epidemiology plus dose-response, not an RCT of smoking assignment.

EFFECT SIZE AND PERSISTENCE

Birthweight deficits are typically on the order of 150–250 g for active smoking in older syntheses, larger at higher consumption. Persistence of size difference into later life is mixed and confounded; do not carry the birthweight effect into IQ.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: partially confounded (20–50%) (maternal smoking during pregnancy shows partial attenuation in sibling and Mendelian randomization studies).

SES, co-exposures, partner smoking, nutrition, residual confounding. Genetically informed designs for this specific path were not in the Chapter 2 dossier.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean a smaller baby has a lesser future, or that smoking is the only growth constraint.

WHAT ACTION IS REASONABLY JUSTIFIED

Population and clinical support for smoking cessation in pregnancy is justified on birthweight/preterm grounds. Treating birth size as destiny is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Kramer (1987), Bulletin of the World Health Organization

Preschool & Early Interventions: Perry, Abecedarian, and Fade-Out at Scale

Two small, high-fidelity 1960s-70s RCTs (Perry, Abecedarian) show large, persistent adult gains; modern at-scale preschool interventions often show cognitive fade-out.

ODERATE-HISTORICAL; LOW-GENERALIZATION | STATUS: DRAFT | DOMAIN: EARLY-LEARNING, COGNITION, POLICY | REPLICATION

WHAT WE KNOW (Narrowest Supported Conclusion)

Two small but rigorously randomized 1960s-70s preschool programs for disadvantaged children (Perry Preschool, Abecedarian) show sustained gains in adult education, employment, earnings, and health, and reduced crime, decades after the intervention ended -- but these are small, intensive, high-fidelity, historically-specific programs, not evidence that any preschool program at scale today replicates these effects.

EVIDENCE AND CAUSAL ASSESSMENT

- Perry Preschool and Abecedarian were both RCTs, each serving roughly 60 children, an unusually strong causal design for early-education research given the era.

- Long-run follow-up (Garcia et al., 2021) found increased adult education, employment, earnings, and health, and lower crime/incarceration in treatment vs. control groups; estimated benefits totaled six-to-ten times program costs.

- Reading/math gains persisted or grew through second grade and were maintained to ages 14-21 in Abecedarian follow-ups (Campbell et al., 2012).

- Perry Preschool evidence also suggests intergenerational spillover: better family environments and outcomes for children of the original participants (Garcia et al., 2023).

- A separate synthesis raises the question of whether modern, larger-scale preschool programs are becoming less effective than these historical small-N demonstration programs, implying scale/fidelity dependence.

EFFECT SIZE AND PERSISTENCE

Effects are large in the original small samples and unusually persistent -- among the longest-followed early-childhood RCTs in the literature (into the 4th-5th decade of life).

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (historical randomized controlled educational trials, not an observational parenting phenotype).

Small sample sizes (~60 per program) limit precision and generalizability; both programs were unusually intensive, high-fidelity, researcher-designed model programs, not typical at-scale public preschool. Benefits are plausibly tied to program quality/intensity/fidelity rather than preschool as a generic category.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not establish that any preschool program, or preschool attendance in general, produces these effect sizes; does not license extrapolating a specific dollar-benefit multiplier without similar fidelity and intensity.

WHAT ACTION IS REASONABLY JUSTIFIED

Institutional investment in high-fidelity, well-resourced early-childhood programs for disadvantaged children is supported by unusually strong, long-run causal evidence from these two programs; assuming any preschool expansion will replicate Perry/Abecedarian-scale effects is not justified without attention to implementation quality.

VERIFIED SOURCES & DOI IDENTIFIERS

- García, Heckman & Ronda (2023), Journal of Political Economy, Perry through late midlife (2021 NBER working paper: w29057)
- Campbell et al. (2012), Developmental Psychology, Abecedarian age-30 follow-up
- Elango, García, Heckman & Hojman (2016), NBER, early childhood education synthesis

Educational Claims About Pretend Play Exceed the Evidence

Pretend play is a real developmental activity with plausible cognitive demands; evidence does not establish that it uniquely causes intelligence, executive function, theory of mind, or creativity.

I-OVERCLAIM; MODERATE-DEMANDS; LOW-UNIQUE-CAUSE | STATUS: DRAFT | DOMAIN: PLAY, EXECUTIVE-FUNCTION | REPLICATIO

WHAT WE KNOW (Narrowest Supported Conclusion)

Pretend play has been claimed to be crucial for healthy development. Lillard et al. (2013) reviewed experimental and correlational evidence for language, narrative, emotion regulation, theory of mind, social skills, and executive function. They concluded that the evidence does not support strong causal claims about unique importance, and that equifinality—multiple routes to similar outcomes—is a serious alternative. Pretend play can recruit dual representation, inhibition of literal reading, memory, flexibility, and language. Those demands do not prove unique causation. Commercial guidelines such as Montessori shelf rotation or toy rotation systems similarly claim enhanced focus and executive development; while minimizing clutter is practical, randomized trials do not establish that formal shelf rotation schedules causally accelerate cognitive milestones.

EVIDENCE AND CAUSAL ASSESSMENT

Lillard et al. (2013) is a systematic narrative review of a large, uneven literature, including null and nonreplicated experimental findings and masked-assessor cautions. Association between pretense and EF is compatible with bidirectionality and selection. Guided-play curricula and landmark preschool RCTs (preschool-early-education-rct.md) are different packages; they are not “pretend play” as such. Chapter 11 play-as-EF remains Low-to-Moderate on that dossier; this card holds the causal claim down.

EFFECT SIZE AND PERSISTENCE

No stable, generalizable effect estimate currently justifies claiming durable, cross-domain gains from pretend play. Short-term task bumps under particular story conditions should not be generalized.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Children with stronger language, EF, and adult support both pretend more complexly and score higher on developmental tasks. Definitions of “play” vary. Publication bias and small samples are live.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Pretend play is not irrelevant. The rejected claim is unique necessity and general cognitive enhancement. Child-led play is not proven EF training. More scheduled “educational play” is not automatically better.

WHAT ACTION IS REASONABLY JUSTIFIED

Protect time and materials for imaginative activity without turning play into a performance target. Using play as a universal intelligence or ToM prescription does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Lillard et al. (2013), Psychological Bulletin

Preterm Birth Raises Group-Level Cognitive Risk, Not a Sentence

A 2024 meta-analysis finds preterm birth associated with lower global cognitive scores (SMD about −0.57); the difference is a population risk, concentrated among the most immature and those with pulmonary morbidity, not an individual forecast.

GRADE: MODERATE-GROUP; LOW-INDIVIDUAL | STATUS: DRAFT | DOMAIN: BIRTH | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Being born before term exposes an infant to extrauterine life before some systems are ready. Across decades of standardized testing, children born preterm score lower, on average, than term-born peers on global cognitive measures. The gap is larger with more extreme immaturity and with neonatal pulmonary morbidity. Contemporary neonatal care is associated with early improvements that, in this synthesis, are not clearly retained after about age 2–3.

EVIDENCE AND CAUSAL ASSESSMENT

Burstein, Aryeh & Geva (2024): 161 studies, 39,799 children; global cognition SMD −0.57 (95% CI −0.63 to −0.52), about 7–9 developmental/IQ points. Observational comparison of preterm vs term; not an experiment that assigns gestational age. Examiner blinding was associated with poorer preterm scores, suggesting possible “compassion bias” when testers know history.

EFFECT SIZE AND PERSISTENCE

SMD ≈ −0.57 is an average. Moderate-to-late preterm cohorts do not show the same later widening as births before 32 weeks in this analysis. Early care-related gains that fade after toddlerhood are a persistence finding, not a reason to treat early scores as destiny.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (clinical cohort meta-analysis of gestational age, not an observational parenting phenotype).

Survival/referral bias; historical treatment change; confounding by indication; attrition. Corrected age matters in early assessment (clinical follow-up guidance such as NICE NG72 is adjacent policy, not this card’s identification).

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Prematurity is not a developmental sentence. A group difference is not a child’s worth, personality, or a parental verdict.

WHAT ACTION IS REASONABLY JUSTIFIED

Risk-sensitive follow-up and family support. Using gestational age as a label that replaces the child is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Burstein, Aryeh & Geva (2024), Developmental Psychology (PMID 39480317)

Early Rank-Order Continuity Is Modest, Not Destiny

Personality and temperament show modest rank-order consistency in early childhood that increases with age; a three-year-old's relative standing is not adult personality.

ERATE-TO-HIGH-PATTERN; LOW-INDIVIDUAL-FORECAST | STATUS: DRAFT | DOMAIN: DEVELOPMENT, TEMPERAMENT | REPLICATI

WHAT WE KNOW (Narrowest Supported Conclusion)

Individual differences persist somewhat. They do not lock. Roberts & DelVecchio (2000) meta-analyzed 152 longitudinal studies (3,217 test–retest correlations) of personality trait rank-order consistency from childhood through old age. Consistency is an increasing function of age. The childhood interval is modest (population estimate about r = .31 for birth to 2.9 years), far below later-adulthood estimates.

EVIDENCE AND CAUSAL ASSESSMENT

This is a quantitative review of test–retest r, not an RCT of "early personality." Measurement changes across ages (heterotypic continuity). The existing temperament card (temperament-reactivity-not-destiny.md) is about dimensional structure, not this age function. Adjacent: Watts delay is a different overclaim (marshmallow as school-success law).

EFFECT SIZE AND PERSISTENCE

An early-childhood r near .3 leaves most variance in relative standing open to later reallocation. Stability coefficients typically rise after early childhood; that rise is not proof that age three was already the adult trait.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (meta-analytic psychometric test-retest stability, not a causal exposure).

Western, questionnaire-heavy samples are common. Informant and interval length affect r. Extreme-tail trajectories can be more persistent than the population average without becoming destiny.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Age three does not determine adulthood. Moderate continuity is not a license to ignore early difficulty, and it is not a license to treat a toddler's standing as a finished self.

WHAT ACTION IS REASONABLY JUSTIFIED

Treat early differences as probabilities that still move. Ranking children by a three-year-old trait score does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Roberts & DelVecchio (2000), Psychological Bulletin

Responsive Caregiving and Attachment Interventions

Randomized sensitivity-training interventions reliably raise caregiver sensitivity and, to a lesser degree, infant attachment security.

ATE-TO-HIGH-SENSITIVITY; MODERATE-ATTACHMENT | STATUS: DRAFT | DOMAIN: ATTACHMENT, PARENTING, SOCIAL | REPLICA

WHAT WE KNOW (Narrowest Supported Conclusion)

Randomized interventions that train parental sensitivity produce a medium-sized increase in sensitive caregiving and a smaller but real increase in infant attachment security; short, focused, behaviorally-oriented interventions outperform longer or more diffuse ones.

EVIDENCE AND CAUSAL ASSESSMENT

- Bakermans-Kranenburg, van IJzendoorn & Juffer (2003) meta-analysis: 70 studies, 88 intervention effects (sensitivity N=7,636; attachment N=1,503). Randomized interventions: d=0.33 for sensitivity, d=0.20 for attachment security.

- Earlier van IJzendoorn et al. (1995) meta-analysis (16 studies, N=869) found d=.58 for sensitivity change and d=.17 for attachment security change; short-term preventive interventions outperformed longer/more intensive/therapeutic approaches.

- De Wolff & van IJzendoorn (1997) meta-analysis of parental antecedents of attachment (66 studies, N=4,176) established sensitivity as a moderate, non-exclusive predictor of attachment security.

EFFECT SIZE AND PERSISTENCE

Sensitivity change: d~0.33-0.58 (medium). Attachment security change: d~0.17-0.20 (small). These are intervention (efficacy) effects from mostly short-term RCTs; long-run persistence into middle childhood and beyond is not established by this literature pass.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (randomized sensitivity intervention trials, not an observational population phenotype).

Moderator analyses consistently show shorter, more behaviorally focused interventions outperform longer/more clinical ones. Effectiveness in multi-problem/high-risk families appears comparable to lower-risk families, suggesting some scalability, but effectiveness trials and cross-cultural replication outside Western/high-income settings are less represented.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not establish that insensitive parenting is a dominant determinant of child outcomes, nor that all attachment insecurity is parent-caused.

WHAT ACTION IS REASONABLY JUSTIFIED

Short, focused, behaviorally-oriented sensitivity-training programs are a reasonably well-supported intervention target for services aiming to support caregiving quality, particularly in higher-risk families; this does not justify blanket judgments about individual parenting style.

VERIFIED SOURCES & DOI IDENTIFIERS

- Bakermans-Kranenburg, van IJzendoorn & Juffer (2003), Psychological Bulletin
- van IJzendoorn, Juffer & Duyvesteyn (1995), J Child Psychol Psychiatry
- De Wolff & van IJzendoorn (1997), Child Development

Feeding as Reciprocal Regulation, Not a Transfer of Calories

Responsive feeding is a cue-based, bidirectional process inside ordinary caregiving; it is not a weight or IQ program, and it is not the suck–swallow or breastfeeding-cognition claim.

DE: MODERATE-FRAMEWORK; LOW-GROWTH-PROGRAM | STATUS: DRAFT | DOMAIN: REGULATION, FEEDING | REPLICATION: VERI

WHAT WE KNOW (Narrowest Supported Conclusion)

After placental transfer ends, feeding is one of the first problems an infant and caregiver solve together. The infant signals hunger, satiety, interest, or refusal. The adult offers, waits, pauses, or persists. Black & Aboud (2011) describe responsive feeding as a four-step reciprocal process: the adult provides structure and an emotional context; the child signals; the adult responds promptly in a contingent, developmentally fitted way; the child meets a predictable consequence. The construct comes from nutrition, psychology, and anthropology. It is a description of an interaction, not a calorie count.

EVIDENCE AND CAUSAL ASSESSMENT

Black & Aboud (2011), Journal of Nutrition: theoretical synthesis for a responsive-feeding supplement, not an RCT of ordinary household differences. Identification is conceptual and review-level. Suck–swallow–breathe coordination (Lau 2015) is a maturational physiology claim. PROBIT/sibling breastfeeding-IQ is a cognition claim. Neither is this card.

EFFECT SIZE AND PERSISTENCE

No portable effect size for ordinary cue-based feeding as a developmental intervention. Persistence of growth or cognitive advantage is not established here.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (behavioral feeding framework, not an observational population phenotype association).

Poverty, food availability, illness, and caregiver strain are easily mistaken for “feeding style.” Cross-cultural feeding practices vary; the four-step description should not be read as a single correct script.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean forcing intake, ignoring satiety, or ranking families by feeding method. Does not mean breastfeeding associations with later cognition. Does not mean a missed feed writes a life.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary noticing of hunger and satiety cues, and skilled feeding support where wanted or needed. Treating a feeding protocol as cognitive enhancement is not justified.

VERIFIED SOURCES & DOI IDENTIFIERS

- Black & Aboud (2011), Journal of Nutrition (DOI 10.3945/jn.110.129973)

Family Routines, Predictability, and Adult Modeling

Predictable family routines and consistent adult behavioral modeling provide proximal scaffolding that supports toddler self-regulation and sleep; routines function as supportive environmental structure, not as a rigid causal machine that installs compliance or conscience.

RATE-SCAFFOLDING; LOW-DIRECT-CAUSAL-CONSCIENCE | STATUS: DRAFT | DOMAIN: PARENTING, SELF-REGULATION | REPLICAT

WHAT WE KNOW (Narrowest Supported Conclusion)

Predictable daily household routines (regular patterns for meals, transitions, and bedtime) and observational adult modeling provide structural scaffolding for early child regulation. By creating a predictable temporal order, routines reduce the cognitive and emotional burden of anticipating transitions. In experimental trials targeting specific routines (such as nightly bedtime sequences), instituting a consistent pattern reliably improves sleep latency and reduces bedtime resistance. However, routines function as proximal environmental scaffolding rather than an autonomous causal mechanism that guarantees long-term self-discipline or moral conscience.

EVIDENCE AND CAUSAL ASSESSMENT

Fiese et al. (2002), Journal of Family Psychology (review of 50 years of research on family routines and rituals across 32 publications), demonstrated that predictable daily routines are robustly associated with positive family cohesion, child emotional well-being, and academic readiness. Crucially, the authors emphasized that routines represent observable family practices embedded within broader emotional climates; observational correlations cannot be interpreted as evidence that isolated routine sequences directly cause positive child traits.

Mindell et al. (2009), Sleep, conducted a randomized controlled trial of 405 infants and toddlers (ages 7 to 36 months) testing a consistent three-step nightly bedtime routine (bath, massage/lotion, quiet activities). Children assigned to the bedtime routine exhibited statistically significant reductions in sleep-onset latency, fewer night wakings, and improved maternal mood compared to controls. This confirms that specific behavioral routines have causal efficacy for immediate situational transitions. It does not demonstrate that routines install enduring, generalized self-control across unrelated domains.

EFFECT SIZE AND PERSISTENCE

In randomized trials of targeted bedtime routines, improvements in sleep latency and nocturnal awakenings show moderate effect sizes ($d \sim 0.40–0.50$) over the intervention window (3 to 6 weeks). In broader observational cohorts, correlations between overall household routine predictability and toddler self-regulation are modest ($r \sim 0.20–0.30$), reflecting immediate behavioral facilitation rather than permanent character formation.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: untested-in-GI (household consistency correlates with parental conscientiousness, executive functioning, and mental health; specific causal attenuation percentages from twin/sibling designs are untested on this card).

Observational studies frequently confound household routines with economic resources and maternal stress: households experiencing financial precarity, erratic work schedules, or housing instability face objective barriers to routine consistency that independently impact child development.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean families must adhere to an inflexible, military-style timetable. Does not mean an occasional disrupted day, family travel, or spontaneous evening harms a child's development. Does not mean children who experience tantrums or behavioral struggles come from disordered or neglectful homes.

WHAT ACTION IS REASONABLY JUSTIFIED

Caregivers can establish predictable, calming sequences around challenging daily transitions—especially bedtime, morning departures, and mealtime—while keeping routines adaptable and warm. Adult calm during transitions provides the primary model from which children learn self-soothing and co-regulation.

VERIFIED SOURCES & DOI IDENTIFIERS

- Fiese et al. (2002), Journal of Family Psychology (DOI: 10.1037/0893-3200.16.4.381 (https://doi.org/10.1037/0893-3200.16.4.381))
- Mindell et al. (2009), Sleep (DOI: 10.1093/sleep/32.5.599 (https://doi.org/10.1093/sleep/32.5.599))

Supine Safe Sleep, Room-Sharing Duration, and Sleep Environment

Placing infants on their backs on a firm, clear surface sharply reduces SIDS mortality, whereas extending room-sharing past 4 to 6 months shows lower-certainty benefits and is associated with shorter consolidated night sleep in the INSIGHT cohort.

GRADE: HIGH | STATUS: DRAFT | DOMAIN: SLEEP, HEALTH | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

High-certainty epidemiological evidence shows that placing an infant wholly on the back (supine) on a firm, flat, non-inclined sleep surface free of soft bedding, pillows, and bumpers—and avoiding surface bed-sharing when a parent is exhausted, smoking, or impaired—substantially reduces the risk of Sudden Infant Death Syndrome (SIDS) and accidental suffocation, with peak vulnerability concentrated between 1 and 4 months of age. By contrast, while room-sharing without bed-sharing in the first 6 months is protective, prospective cohort evidence (the INSIGHT study) suggests that continued room-sharing at 4 and 9 months is associated with less consolidated night sleep, more frequent night feedings, and higher odds of transitioning into reactive bed-sharing overnight.

EVIDENCE AND CAUSAL ASSESSMENT

- AAP 2022 Evidence Synthesis (moon-2022-sids): Synthesizes case-control and population surveillance studies showing that prone (stomach) and side sleeping increase SIDS odds roughly 2- to 13-fold, particularly when infants accustomed to supine sleep are placed prone unaccustomed ("unaccustomed prone"). Once an infant can independently roll from back to stomach and stomach to back (typically 4–6 months), the infant does not need to be repositioned if the crib is completely clear of soft objects and swaddling has been discontinued.

- INSIGHT Prospective Cohort (paul-2017-insight, N=230 mother-infant dyads): Infants who slept in their own room by 4 months or between 4 and 9 months obtained longer longest-sleep stretches (+45 to +46 minutes at 9 months) and more total night sleep at 4, 9, and 30 months than infants who remained in the parents' room at 9 months, and were four times less likely to be brought into the adult bed overnight.

EFFECT SIZE AND PERSISTENCE

Population-wide "Back to Sleep" campaigns reduced SIDS rates by more than 50% across high-income nations in the 1990s (moon-2022-sids). In the INSIGHT cohort (paul-2017-insight), independent sleepers at 9 months slept an average of 10 hours and 37 minutes at night versus 9 hours and 47 minutes for room-sharers, with differences in bedtime routines and night-sleep duration still detectable at 30 months.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: partially confounded (20–50%) for room-sharing duration and sleep consolidation (paul-2017-insight), as parents of more wakeful infants may delay moving them to a separate room (reverse causation), though the INSIGHT study adjusted for baseline infant fussiness and feeding mode. By contrast, the mechanical and physiological hazard of prone sleep on soft bedding in early infancy is robust (<20%).

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean parents must wake up all night to flip a 6-month-old who has rolled onto their stomach on their own in an empty crib, provided arms-in swaddling has been stopped. It also does not mean moving an infant to a nursery at 5 or 6 months is reckless when all core safe-sleep rules (back to sleep, firm flat mattress, empty crib, smoke-free home) are followed.

WHAT ACTION IS REASONABLY JUSTIFIED

Always place infants on their backs on a firm, flat, empty crib or bassinet for every sleep, stop swaddling as soon as the infant shows signs of attempting to roll (typically 8–12 weeks), share a room (on a separate surface) for at least the first 4 to 6 months when SIDS risk is highest, and treat the transition to a separate nursery after 6 months—and the use of low-volume (<50 dBA) white noise placed across the room—as pragmatic household choices rather than moral tests.

VERIFIED SOURCES & DOI IDENTIFIERS

- Moon, Carlin & Hand (2022), Pediatrics, AAP evidence report on sleep-related infant deaths (moon-2022-sids)
- Paul et al. (2017), Pediatrics, INSIGHT study on mother-infant room-sharing and sleep (paul-2017-insight)

Screen Time and Language Development

Screen exposure is a decomposable, mostly observational predictor of language: quantity is negatively associated, co-viewing and educational content buffer.

GRADE: LOW-TO-MODERATE | STATUS: DRAFT | DOMAIN: LANGUAGE, ECOLOGY, PARENTING | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Screen time is not a unitary exposure: total quantity of screen use is negatively associated with language skills in early childhood, but co-viewing with an adult, educational content, and later onset age are each associated with neutral-to-positive language outcomes.

EVIDENCE AND CAUSAL ASSESSMENT

- Madigan et al. (2020) meta-analysis, JAMA Pediatrics (42 studies, N=18,905, ages 0-12): screen time negatively associated with language (r=-0.14); background TV more negatively associated (r=-0.19); educational programs (r=+0.13) and co-viewing (r=+0.16) positively associated; later onset age positively associated (r=+0.17).

- Danish TRACES survey (N=31,125 toddlers): mobile device screen time >=1hr/day associated with higher odds of language comprehension difficulties (AOR 1.30-1.42) and expressive difficulties (AOR 1.19-1.46); frequent reading partially buffered comprehension effects.

- A separate meta-analysis (28 studies, ages 0-3, 2024) found screen time negatively related to language, co-viewing and later onset positively associated, educational content non-significant.

- Jing, Ye, Kirkorian & Mares (2023), Child Development (distinct from Madigan 2020): overall small positive association between screen exposure and vocabulary (r=.23) across 0–6 years, illustrating heterogeneity by outcome/measurement approach. Not a retraction of Madigan 2020.

EFFECT SIZE AND PERSISTENCE

Small effect sizes throughout (r's typically 0.10-0.20); dose-response patterns reported (>2h/day associated with larger deficits). Persistence beyond toddlerhood/preschool is not well established in this literature pass.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: untested-in-GI (screen exposure observational associations correlate with parental education, household chaos, and shared genetic traits; not yet tested in within-family sibling-comparison or twin designs).

Almost all evidence is observational; most measurement is parent-reported total duration, which the project's own standards classify as weak causal evidence. Reverse causation and confounding by SES/parental engagement are both plausible and largely unaddressed. Content, context, and co-use are decomposed in some but not all studies. Madigan 2020 (language skills, quantity negative) and Jing 2023 (vocabulary, overall positive) are heterogeneous outcome packages, not a failed identification of a single paper.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not establish that screens categorically or causally damage language development, nor that any specific duration threshold is a validated causal cutoff.

WHAT ACTION IS REASONABLY JUSTIFIED

Given consistent (if weak-to-moderate causally) associations concentrated in passive/solo/background use, and consistent buffering by co-viewing and educational content, proportionate guidance favors decomposing screen exposure over blanket duration limits -- but strong prescriptive claims are not justified by current evidence quality.

VERIFIED SOURCES & DOI IDENTIFIERS

- Madigan, McArthur, Anhorn, Eirich & Christakis (2020), JAMA Pediatrics (DOI 10.1001/jamapediatrics.2020.0327; PMC7091394)
- Jing, Ye, Kirkorian & Mares (2023), Child Development (DOI 10.1111/cdev.13927)
- TRACES cohort, BMC Public Health (2024)
- Screen exposure and early language development, meta-analysis (2024)

Self-Regulation and Delay of Gratification (Marshmallow Test)

A large NICHD-cohort replication shows the original marshmallow-test predictive claim is mostly explained by SES and early cognitive ability.

O-HIGH-OVERSTATEMENT-OF-ORIGINAL | STATUS: DRAFT | DOMAIN: SELF-REGULATION, EXECUTIVE-FUNCTION, COGNITION | RE

WHAT WE KNOW (Narrowest Supported Conclusion)

Early delay-of-gratification ability predicts later achievement only weakly once family socioeconomic status, home environment, and early cognitive ability are statistically controlled, contradicting the original narrative that early self-control is a strong, independent predictor of life outcomes.

EVIDENCE AND CAUSAL ASSESSMENT

- Watts, Duncan & Quan (2018), Psychological Science: conceptual replication using N=918 children from the NICHD SECCYD cohort, larger and more diverse than Mischel's original ~90-child sample.

- An additional minute of delay at age 4 predicted only ~0.1 SD gain in achievement at age 15 in the full sample; predictive power dropped to near zero after controlling for SES, home environment, and early cognitive ability.

- No association was found between delay ability and later behavioral or personality measures once covariates were included.

- A preregistered 2024 replication (N=702) similarly found the test does not reliably predict adult functioning.

EFFECT SIZE AND PERSISTENCE

Original correlational claims implied a strong, durable trait-like effect; the 2018 replication's controlled effect size (beta~0.07) is small and much of the raw bivariate association is explained by family SES and early cognitive/home-environment differences.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: largely confounded (>50%) (marshmallow delay associations with long-term achievement largely attenuate after controlling for early ability and family background).

The original Mischel studies used a small, homogeneous (Stanford-affiliated preschool) sample. Family instability and environmental trust plausibly explain some children's rational choice to take the immediate reward rather than reflecting a stable deficit in self-regulation capacity.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not support claims that a child's early performance on a delay task fixes or predicts their life trajectory, nor that narrow delay-of-gratification interventions will meaningfully change long-term outcomes.

WHAT ACTION IS REASONABLY JUSTIFIED

Framing self-regulation as fixed or destiny-determining in early childhood is not supported; addressing family economic stability and environmental predictability is better supported than narrow willpower training approaches.

VERIFIED SOURCES & DOI IDENTIFIERS

- Watts, Duncan & Quan (2018), Psychological Science, NICHD SECCYD replication
- Mischel, Shoda & Peake (1990), Developmental Psychology (historical claim only)

Severe Institutional Deprivation Is Not Ordinary Parenting

Randomized foster-care placement after early institutional care improves some cognitive and relational outcomes; the contrast is severe deprivation, not typical variation in sensitivity or daycare.

MODERATE-TO-HIGH-PLACEMENT; VERY-LOW-ORDINARY-CARE | STATUS: DRAFT | DOMAIN: ATTACHMENT, REGULATION, SOCIAL |

WHAT WE KNOW (Narrowest Supported Conclusion)

Severe early institutional deprivation can impair cognitive, socioemotional, and physical development. In the Bucharest Early Intervention Project, abandoned children already living in institutions were randomly assigned to high-quality foster care or care as usual. Family placement improved cognitive outcomes relative to remaining institutionalized; recovery was incomplete and depended on timing and stability. This is a deprivation-versus-family-care contrast, not a study of ordinary parental inconsistency.

EVIDENCE AND CAUSAL ASSESSMENT

Nelson, Zeanah, Fox et al. (2007), Science: 136 institutionalized children randomized; cognitive outcomes through 54 months were higher in the foster-care arm, especially among the youngest placed. A later overview (Nelson, Fox & Zeanah, 2023/2024) summarizes follow-ups through adolescence: benefits in several domains, incomplete recovery, and timing effects that the authors describe as possible sensitive periods for this deprivation package. Random assignment is unusually strong for this question; the Romanian institutional conditions of the trial (rotating caregivers, low stimulation, bundled health and nutritional risk) do not represent ordinary parenting.

EFFECT SIZE AND PERSISTENCE

Group differences are large relative to never-institutionalized community comparisons and smaller, still meaningful, for foster care versus care as usual. Persistence is domain-specific; some advantages remain, others do not fully close. Placement stability matters. These are not portable “attachment treatment” effect sizes.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (randomized foster-placement intervention following severe deprivation, not ordinary parenting).

The intervention is a package (stable caregivers, stimulation, nutrition, health, material support). Isolating “attachment” from the rest is not licensed. Attrition and later placement changes complicate long follow-up. Ethical and historical uniqueness limits generalization.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Ordinary parental mistakes, ordinary daycare, or ordinary variation in sensitivity do not equal institutional deprivation. Early adversity does not make recovery impossible. The Strange Situation is not a personality test derived from this trial.

WHAT ACTION IS REASONABLY JUSTIFIED

Prevent severe neglect and unnecessary institutionalization; provide stable family-based care. Using BEIP to rank typical parents is not justified.

VERIFIED SOURCES & DOI IDENTIFIERS

- Nelson, Zeanah, Fox, Marshall, Smyke & Guthrie (2007), Science 318:1937–1940
- Nelson, Fox & Zeanah (2023/2024), Current Directions in Psychological Science (PMC10977996)

Early Skin-to-Skin Contact: Immediate Physiology, Not Character

In healthy newborns, early skin-to-skin contact is associated with higher breastfeeding rates and some short-term physiologic markers; GRADE is mixed, and the practice is not a lifelong character intervention.

DERATE-BREASTFEEDING; LOW-INFANT-PHYSIOLOGY | STATUS: DRAFT | DOMAIN: BIRTH, FEEDING, REGULATION | REPLICATION

WHAT WE KNOW (Narrowest Supported Conclusion)

Early skin-to-skin contact (naked infant prone on the mother's chest soon after birth) is a bundled care practice, not a single dose of affection. In healthy, mostly term newborns, Cochrane synthesis finds more breastfeeding at one to four months and mixed, lower-certainty infant physiologic results. Kangaroo mother care for low-birth-weight infants is a different population and must not be averaged into this card.

EVIDENCE AND CAUSAL ASSESSMENT

Moore, Bergman, Anderson & Medley (2016): 38 trials with data (3,472 women); most small; none met all good-quality criteria; no trial was blinded. Comparator is usual hospital care. This is an RCT synthesis of a bundled, unblindable intervention. Benefits cannot be assigned to parental love alone when warmth, feeding support, and reduced separation travel together.

EFFECT SIZE AND PERSISTENCE

Breastfeeding maintenance is the strongest identified outcome here. Duration findings are smaller-sample and GRADE low. Infant temperature difference is statistically present and clinically trivial. Long-term neurodevelopment is not established by this review.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (immediate postpartum clinical physiology and RCT trials, not an observational phenotype).

Performance bias is structural (the intervention is visible). Small samples, co-interventions, and heterogeneous protocols. SSC is not KMC; mixing term SSC with mortality results from very-low-birth-weight kangaroo care is a translation error.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean SSC guarantees bonding, breastfeeding, or cognitive advantage, or that its absence is a character injury. It does not license treating contact as mandatory moral performance.

WHAT ACTION IS REASONABLY JUSTIFIED

Make safe, supported contact available as part of essential newborn care. Moralizing contact, or treating a missed hour as a life sentence, is not justified by these trials.

VERIFIED SOURCES & DOI IDENTIFIERS

- Moore, Bergman, Anderson & Medley (2016), Cochrane Database of Systematic Reviews (PMC6464366)

Oral Feeding Requires Suck–Swallow–Breathe Coordination

Safe oral feeding is a maturational coordination of sucking, swallowing, and respiration; the elements mature at different rates, especially in preterm infants, and feeding method is not parental worth.

ADE: HIGH-COORDINATION; MODERATE-PRACTICE-EFFECTS | STATUS: DRAFT | DOMAIN: BIRTH, FEEDING | REPLICATION: VERIFI

WHAT WE KNOW (Narrowest Supported Conclusion)

After birth, nutrition must move from placental transfer to oral feeding. That requires sucking, swallowing, breathing, arousal, and gastrointestinal function to work together. Each function has parts that mature at different times. “Coordination” is therefore not a single switch. Preterm infants often take longer to feed safely by mouth; swallow–respiration matching is a common bottleneck.

EVIDENCE AND CAUSAL ASSESSMENT

Lau (2015), Annals of Nutrition and Metabolism (PMC4530609; author’s neonatology group), reviews nutritive sucking (suction and expression), swallow processing, and respiration, arguing that readiness is not well defined clinically and that nonnutritive sucking is not sufficient evidence of oral-feeding readiness. Designs are physiologic observation and preterm/term comparison, not randomized feeding-method trials for later IQ. breastfeeding-cognitive-development.md remains the place for PROBIT/sibling cognition.

EFFECT SIZE AND PERSISTENCE

The immediate outcome is safe transfer of a bolus. Persistence of feeding difficulty is clinical and heterogeneous. There is no license to read feeding method as a character or intelligence result.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (maturational oral-motor physiology, not an observational population phenotype association).

Gestational age, illness, congenital difference, posture, and caregiver/clinical support are intertwined. Breast, bottle, expressed milk, and tube feeding are distinct situations.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Feeding difficulty is not a verdict on parental worth. Breastfeeding associations with later cognition are a separate, heavily confounded literature.

WHAT ACTION IS REASONABLY JUSTIFIED

Skilled feeding support where wanted or needed, and evaluation of unsafe feeding. Ranking families by feeding method is not justified here.

VERIFIED SOURCES & DOI IDENTIFIERS

- Lau (2015), Annals of Nutrition and Metabolism (PMC4530609)

Symbolic and Pretend Play Emerge Across the Second Year

Object substitution and simple pretense commonly become visible between about 12 and 24 months; this is a representational transition, not a required intelligence program or a diagnostic cutoff.

-TO-HIGH-EMERGENCE; MODERATE-MECHANISM; LOW-DIAGNOSTIC-CUTOFF | STATUS: DRAFT | DOMAIN: PLAY, COGNITION | REPL

WHAT WE KNOW (Narrowest Supported Conclusion)

Play is not one exposure. Belsky & Most (1981) described a cross-sectional sequence from exploratory object manipulation toward more symbolic acts across infancy. During the second year, many children begin treating an object or action as standing for something else. Lillard et al. (2013) take that developmental emergence as real while rejecting unique-necessity claims about later intelligence. Adjacent exploration-as-sampling.md and embodied-perception-action.md cover information-seeking and motor–perception coupling, not pretense.

EVIDENCE AND CAUSAL ASSESSMENT

Observational and cross-sectional studies show increasing nonliteral object use; they do not isolate a single cognitive module as the cause. Emergence is reciprocal with language, memory, social routines, and executive control. This card is not an intervention efficacy claim (see pretend-play-educational-claims.md). Toddler peer coordination is toddler-peer-interaction.md.

EFFECT SIZE AND PERSISTENCE

Age windows are descriptive. Complexity expands with partners, materials, and language. There is no portable effect size for “more pretend.”

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Tasks underestimate children whose play is less verbal or culturally unfamiliar. Prop familiarity and adult scaffolding shape what is coded as pretense.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

A prescribed play sequence is not required. Variation in observed pretend play is not by itself pathology. More toys are not more representation. Play is not a manufactured intelligence program.

WHAT ACTION IS REASONABLY JUSTIFIED

Treat symbolic play as one observation among many. Forcing a script or ranking families by pretense frequency does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Belsky & Most (1981), Developmental Psychology
- Lillard et al. (2013), Psychological Bulletin

Temperament as Reactivity and Control, Not Destiny

Early-emerging differences in reactivity, approach, and attentional control are measurable and moderately stable; they are not a parenting verdict or a fixed personality forecast.

GRADE: MODERATE-DIFFERENCES; LOW-DESTINY | STATUS: DRAFT | DOMAIN: EMOTION, TEMPERAMENT | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Infants differ in how readily they become distressed, how they approach novelty, and how they can be soothed or redirected. Gartstein & Rothbart (2003) used the Revised Infant Behavior Questionnaire to recover a three-factor structure—surgency/extraversion, negative affectivity, and regulatory capacity/orienting—that organizes those differences as dimensions, not types. The Chapter 9 plan treats this as a constitutional contribution to interaction, not a completed character.

EVIDENCE AND CAUSAL ASSESSMENT

The IBQ-R is a caregiver-report instrument with factor structure recovered in a large infant sample. Laboratory observations converge on similar dimensions but are narrower. Rank-order stability is typically moderate, not identity-preserving. Genetically informed designs in the broader temperament literature show both child effects on caregivers and evocative transactions; this card does not claim a parent-to-child causal law. Adjacent: behavioral-inhibition-anxiety-risk.md is a narrower fear/approach claim.

EFFECT SIZE AND PERSISTENCE

Stability coefficients are typically in a moderate range across infancy into toddlerhood. Mean-level change is expected. A high-reactivity infant in a well-matched context is not the same developmental object as the same scores under chronic stress.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (caregiver-report psychometric trait with ~40–60% heritability, not an environmental exposure).

Parent-report is contaminated by expectation and current behavior. Culture shapes which expressions are noticed. Goodness-of-fit and differential susceptibility remain live alternatives to “difficult child.”

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Difficult temperament does not equal poor parenting or poor outcome. High reactivity is not intrinsically maladaptive. Temperament is not a license to withdraw opportunities or to treat the child as a finished type.

WHAT ACTION IS REASONABLY JUSTIFIED

Match demands and support to the child without turning a questionnaire into an identity. Labeling destiny does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Gartstein & Rothbart (2003), Infant Behavior and Development

Discipline and Behavioral Boundaries: Time-Outs, Physical Punishment, and Escalation

Physical punishment is associated with increased child aggression, externalizing problems, and antisocial behavior without promoting compliance; calm, brief time-out protocols serve as effective de-escalation tools without damaging attachment.

ATE-SIBLING-TRIANGULATION; LOW-TECHNIQUE-ALONE | STATUS: DRAFT | DOMAIN: SOCIAL-EMOTIONAL, BOUNDARIES | REPLICA

WHAT WE KNOW (Narrowest Supported Conclusion)

When toddlers engage in dysregulated behaviors such as hitting, biting, kicking, or refusing rules, parental responses strongly shape subsequent aggression trajectories. How to respond to temper tantrums is a core operational challenge: pairing calm adult coregulation with brief non-punitive pauses breaks coercive reinforcement cycles far more effectively than shouting or force. Gershoff & Grogan-Kaylor (2016) synthesized 75 studies (N > 160,000 children) across five decades, finding that physical punishment (spanking) is consistently associated with increased aggressive behavior, externalizing problems, reduced cognitive ability, and weakened moral internalization, with no evidence of superior short- or long-term compliance compared to nonphysical discipline. In contrast, well-implemented brief time-out protocols (1–3 minutes of quiet emotional down-regulation within a warm, responsive relationship) consistently reduce coercive cycles, aggression, and oppositional defiance across randomized controlled trials without adverse effects on attachment security or emotional health (Dadds & Tully 2019).

EVIDENCE AND CAUSAL ASSESSMENT

Meta-analytic evidence from Gershoff & Grogan-Kaylor (2016) specifically separated mild physical punishment (spanking without injury) from severe physical abuse, demonstrating that spanking alone shares the direction and pattern of adverse externalizing associations. Longitudinal fixed-effects and within-family sibling comparison studies attenuate effect sizes but preserve statistically significant associations between spanking and toddler aggression. Dadds & Tully (2019) reviewed empirical evaluations of evidence-based parenting programs (Parent-Child Interaction Therapy, Triple P, Incredible Years), confirming that brief time-out functions as a nonviolent contingency and emotional pause that breaks coercive reinforcement traps.

EFFECT SIZE AND PERSISTENCE

Meta-analytic effect sizes for physical punishment and externalizing aggression average d = 0.20 to $0.35. Across randomized parent-training trials, implementing predictable nonphysical boundaries (including brief time-outs combined with 5:1 positive-to-negative interactions) yields moderate-to-large reductions in toddler physical aggression (d = 0.40to $0.70) that persist at 1- to 3-year follow-ups.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: partially confounded (20–50%). Genetically informed designs and twin differences show that bidirectional child-effects account for roughly a third of the observational association (difficult, aggressive toddlers evoke harsher responses), yet nonphysical discipline remains protective against escalating aggression after accounting for shared genetic confounding.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean a parent who spanked once in acute exhaustion has permanently damaged their toddler. Does not mean toddlers naturally resolve aggressive outbursts without proactive adult intervention and clear physical boundaries. Does not mean time-out should involve isolation, humiliation, screaming, or emotional abandonment; punitive isolation increases fight-or-flight distress rather than self-regulation.

WHAT ACTION IS REASONABLY JUSTIFIED

Parents facing toddler hitting, biting, and boundary testing should immediately eliminate physical punishment and implement nonviolent, predictable boundaries: brief calm pauses (1 minute per year of age), physical containment to protect safety, repair after distress subsides, and proactive reinforcement of constructive substitute behaviors.

VERIFIED SOURCES & DOI IDENTIFIERS

- Gershoff, E. T., & Grogan-Kaylor, A. (2016). Spanking and child outcomes: Old controversies and new meta-analyses. Journal of Family Psychology, 30(4), 453–469. DOI: 10.1037/fam0000191

- Dadds, M. R., & Tully, L. A. (2019). What is it to be a good parent? The role of discipline, warm parenting, and children's conduct problems. Journal of Child Psychology and Psychiatry, 60(9), 1018–1026. DOI: 10.1111/jcpp.13088

Toddler Peer Interaction Is Not a Social-Skills Commodity

In the second and third years, another toddler is a less accommodating partner than a caregiver; peer contexts create coordination demands, but hours of peer exposure do not automatically produce later prosocial competence.

GRADE: MODERATE-CONTEXT; LOW-HOURS-CAUSE | STATUS: DRAFT | DOMAIN: SOCIAL, PLAY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

A caregiver is usually highly motivated to understand and adapt to a child. Another one- or two-year-old is not. Brownell, Ramani & Zerwas (2006) showed that coordinated peer activity (for example, cooperating to work a toy) is limited at one year and becomes more possible across the second year as social understanding and shared goals come online. Early peer life is therefore a context with distinctive demands — imitation, object conflict, repair — not a background object and not a purchased skill.

EVIDENCE AND CAUSAL ASSESSMENT

Laboratory and structured-play observations, not randomization to “more peers.” Child temperament, existing competence, and adult scaffolding select who plays with whom. Later longitudinal links from preschool peer quality to prosocial behavior are transactional and are not this card’s claim.

EFFECT SIZE AND PERSISTENCE

No portable standardized effect for “peer exposure.” The defensible unit is age-typical change in coordinated activity under specified tasks.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Group composition, adult scaffolding, and which child already has more skill all confound “peer effects.” Aggressive or chaotic groups are a different exposure.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean peer exposure automatically improves social skills. Does not mean an only child is deprived. Does not mean stranger wariness is a peer-skills failure. Does not mean group care is institutional deprivation (severe-institutional-deprivation.md).

WHAT ACTION IS REASONABLY JUSTIFIED

Treat safe, reciprocal play with other children as a possible context, not a commodity. Quality, safety, and adult support matter more than manufacturing a peer schedule.

VERIFIED SOURCES & DOI IDENTIFIERS

- Brownell, Ramani & Zerwas (2006), Child Development

Toddler Physical Aggression Usually Peaks and Declines

Physical aggression commonly rises through the second and third years and declines for most children; a smaller group remains elevated, and presence of hitting is not a violence forecast.

TO-HIGH-TRAJECTORY; MODERATE-INDIVIDUAL; LOW-VIOLENCE-FORECAST | STATUS: DRAFT | DOMAIN: BEHAVIOR, SOCIAL | REP

WHAT WE KNOW (Narrowest Supported Conclusion)

Physical hitting, grabbing, and related acts are common in toddlerhood. Tremblay et al. (2004) described trajectories of physical aggression in early childhood: a typical rise and later decline for most children, and a smaller group with persistently elevated frequency. Frequency, severity, context, and change over time are more informative than presence alone.

EVIDENCE AND CAUSAL ASSESSMENT

Longitudinal cohorts identify multiple trajectories. These designs support probabilistic pathways, not inevitability. Child characteristics, language, sibling and peer contexts, caregiver responses, and stress all participate. This is not a genetically informed attenuation package on this card. Parent-training efficacy belongs to Chapter 14 and is not this identification.

EFFECT SIZE AND PERSISTENCE

Most children reduce physical aggression after the toddler peak. Persistence is less common and more consequential. Aggression can change form rather than disappear.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: untested-in-GI (early physical aggression is moderately heritable with environmental moderation across typical cohorts; quantitative sibling/adoption attenuation for trajectory shapes remains untested).

Parent-report and observer measures differ. Sexed rating biases, socioeconomic stress, sleep, disability, and exposure to violence are intertwined with caregiver responses.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Toddler hitting does not predict later violence as destiny. Ordinary aggression should not be ignored when it is severe, dangerous, persistent, or impairing. Aggression is not proof of poor parenting or a failed “no.”

WHAT ACTION IS REASONABLY JUSTIFIED

Developmentally informed monitoring and targeted support for persistent or dangerous patterns. Alarmist labels for common transient behavior do not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Tremblay et al. (2004), Pediatrics

Toilet Learning Readiness and Initiation Timing

Prospective cohort studies show that initiating intensive toilet training before 24 to 27 months lengthens total training duration by 8 to 12 months without advancing the final age of daytime dryness, which depends on physiological and behavioral readiness.

GRADE: MODERATE | STATUS: DRAFT | DOMAIN: MOTOR, SELF-REGULATION | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Prospective pediatric cohort evidence suggests that the age at which a toddler achieves reliable daytime dryness is governed primarily by neurological and physiological maturation—specifically bladder capacity, voluntary external sphincter control, and cognitive-communicative readiness—rather than how early a parent begins training. Initiating intensive toilet training between 18 and 22 months is associated with a much longer training process (often 10 to 15 months of accidents and reminders) while yielding nearly the same final completion age (roughly 32 to 36 months) as waiting until the child shows clear readiness signs around 24 to 28 months.

EVIDENCE AND CAUSAL ASSESSMENT

- *Blum, Taubman & Nemeth (blum-2003-toilet, N=406 children followed prospectively in Pediatrics):* Initiation of intensive toilet training between 17 and 21 months correlated strongly with longer training duration ($r = -0.71$, $p < 0.001$), taking an average of 13 to 14 months to complete, whereas initiating after 27 months took an average of 5 to 6 months, with minimal difference in the final age of completion (~35–36 months) and higher rates of constipation and stool withholding when pressure was applied before readiness.

- Schum et al. (schum-2002-toilet, N=267 toddlers followed longitudinally): Mapped the sequential acquisition of 11 toilet-training readiness skills, showing that median completion of daytime dryness occurs around 32.5 months in girls and 35.0 months in boys, preceded by staying dry for 2 hours, showing interest in the potty, indicating a wet/soiled diaper, and pulling pants up and down.

EFFECT SIZE AND PERSISTENCE

Correlation between earlier initiation age (<27 months) and longer duration of training is large ($r = -0.71$, blum-2003-toilet), translating to 7–9 extra months of active training for almost no gain in the calendar age at which the child is reliably dry. Nighttime dryness (nocturnal continence) matures on an even slower biological clock—driven by nighttime antidiuretic hormone (vasopressin) surges and deep-sleep arousal thresholds—and remains normal to take until ages 4 to 6.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: partially confounded (20–50%) for behavioral resistance outcomes (such as stool toileting refusal), as child temperament (high negative reactivity or sensory sensitivity) influences both when parents start and how strongly toddlers resist coercive schedules.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean parents cannot introduce a potty chair casually in the bathroom at 18 to 20 months so the toddler can sit on it fully clothed or before bath time. It means that treating a 20-month-old's accidents during a rigid "three-day bootcamp" as stubbornness mistakes an immature spinal-frontal bladder circuit for a behavioral choice.

WHAT ACTION IS REASONABLY JUSTIFIED

Watch for concrete readiness signs rather than a calendar birthday—staying dry for two hours at a stretch, recognizing the physical urge to pee or poop, communicating before or during elimination, able to lower and pull up simple elastic pants, and wanting to imitate older family members—and pause immediately if a toddler develops stool withholding or chronic constipation, which turns a biological milestone into a painful power struggle.

VERIFIED SOURCES & DOI IDENTIFIERS

- Blum, Taubman & Nemeth (2003), Pediatrics, Initiation age vs. duration of toilet training (blum-2003-toilet)
- Schum et al. (2002), Pediatrics, Sequential acquisition of toilet-training skills (schum-2002-toilet)

Working Memory and Cognitive Shifting in Infancy and Toddlerhood

The capacity to maintain representations across delay and shift attentional sets develops systematically between 6 months and 3 years, linked to prefrontal maturation; task performance reflects emergent developmental capacity, not a fixed quotient of future intelligence.

TE-TO-HIGH-TRAJECTORY; LOW-INDIVIDUAL-FORECAST | STATUS: DRAFT | DOMAIN: COGNITION, EXECUTIVE-FUNCTION | REPLIC

WHAT WE KNOW (Narrowest Supported Conclusion)

Executive function is not a unitary skill that arrives all at once. Working memory (the ability to hold mental representations active across a temporal delay to guide subsequent action) and cognitive shifting (the capacity to disengage from an established rule set and switch attention to a new dimension) follow marked developmental trajectories across the first three years of life. Performance on delayed search tasks (such as Piaget's A-not-B paradigm) improves progressively between 7.5 and 12 months, reflecting functional maturation of the dorsolateral prefrontal cortex. Dimensional task switching advances substantially between ages 3 and 4, transitioning from perseverative rule maintenance to flexible set-shifting.

EVIDENCE AND CAUSAL ASSESSMENT

Diamond (1985), Child Development, longitudinally tracked infants aged 6 to 12 months on the A-not-B task with varying delay intervals (0 to 10 seconds). The duration of delay infants could tolerate before committing the perseverative reaching error increased systematically at roughly 2 to 3 seconds per month between 7.5 and 12 months. Successful search required the co-maturation of two interrelated capacities: working memory to maintain the representation of the hidden object's new location (B) across the delay, and inhibitory control to suppress the conditioned motor habit of reaching to the previously rewarded location (A).

Zelazo (2006), Nature Protocols, systematized the Dimensional Change Card Sort (DCCS) paradigm, demonstrating that 3-year-old children can readily sort cards by an initial dimension (e.g., color) and can verbally state the post-switch rule (e.g., sort by shape), but nevertheless perseverate by continuing to sort by the original dimension. By age 4 to 5, children reliably shift between dimensions, reflecting the development of higher-order hierarchical rule representation (the Cognitive Complexity and Control theory).

EFFECT SIZE AND PERSISTENCE

Normative developmental transitions exhibit large age-related effects ($d > 0.8$ between 7 and 12 months on A-not-B delay duration, and between age 3 and age 4 on post-switch DCCS sorting). However, individual rank-order stability across early childhood is modest ($r \sim 0.30–0.40$); infant working-memory metrics share substantial variance with general processing speed and receptive vocabulary, rather than acting as an isolated, stable executive trait.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: untested-in-GI (normative developmental trajectory replicated across international cohorts; genetic and environmental contributions to early infant delay capacity are largely shared with general cognitive ability and linguistic comprehension).

Task-specific confounding is pronounced: an infant's failure on delayed reaching can arise from transient motor clumsiness, poor spatial acuity, fussiness, or examiner distraction rather than cognitive incapacity. In toddler card sorts, failure to switch often reflects attentional capture by salient perceptual features rather than a general absence of memory.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean an 8-month-old who searches at location A lacks memory or object permanence. Does not mean a 3-year-old who perseverates on the card sort is stubborn or cognitively delayed. Does not support commercial claims that infant flashcards, digital apps, or structured drill exercises will build a superior working memory.

WHAT ACTION IS REASONABLY JUSTIFIED

Caregivers should provide rich everyday opportunities for predictable, playful engagement that exercises working memory and cognitive flexibility—such as peekaboo, hide-and-seek, simple sorting during cleanup, and songs with alternating actions—while recognizing that perseveration and short-term forgetting are normal signatures of developing prefrontal circuits.

VERIFIED SOURCES & DOI IDENTIFIERS

- Diamond (1985), Child Development (DOI: 10.2307/1130099 (https://doi.org/10.2307/1130099))
- Zelazo (2006), Nature Protocols (DOI: 10.1038/nprot.2006.46 (https://doi.org/10.1038/nprot.2006.46))